COMPUTER NETWORKING



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Networking

- **Computer network** A collection of computing devices that are connected in various ways in order to communicate and share resources

  Usually, the connections between computers in a network are made using physical wires or cables

  However, some connections are **wireless**, using radio waves or infrared signals

A network consists of 2 or more computers connected together, and they can communicate and share resources (e.g. information)

A network is a set of devices (often referred to as nodes) connected by communication link for the purpose of sharing resources. A **node** can be a computer, router, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

PURPOSE OF COMPUTER NETWORKS
WHY NETWORK COMPUTERS ?

- Remote data Access e.g. Remote offices & branches
- Sharing Hardware e.g. Printers, Scanners etc
- Sharing Software e.g. All staff using one Software located on the Server
- Effective and Efficient Communication e.g. skype, twitter etc
- Sharing Storage through a Centralised Storage Server
- Effective File Sharing

Network Evaluation Criteria

- **Performance**
  - Mostly measured by throughput and delay
- **Reliability**
  - Is the Network able to provide the resources it Promises to provide ?
  - The frequency of failure
  - Recovery time from a failure
- **Security**
  - Protecting data from
    - unauthorized access
    - Damage
- **Availability**

Is the Network accessible as and when required ?

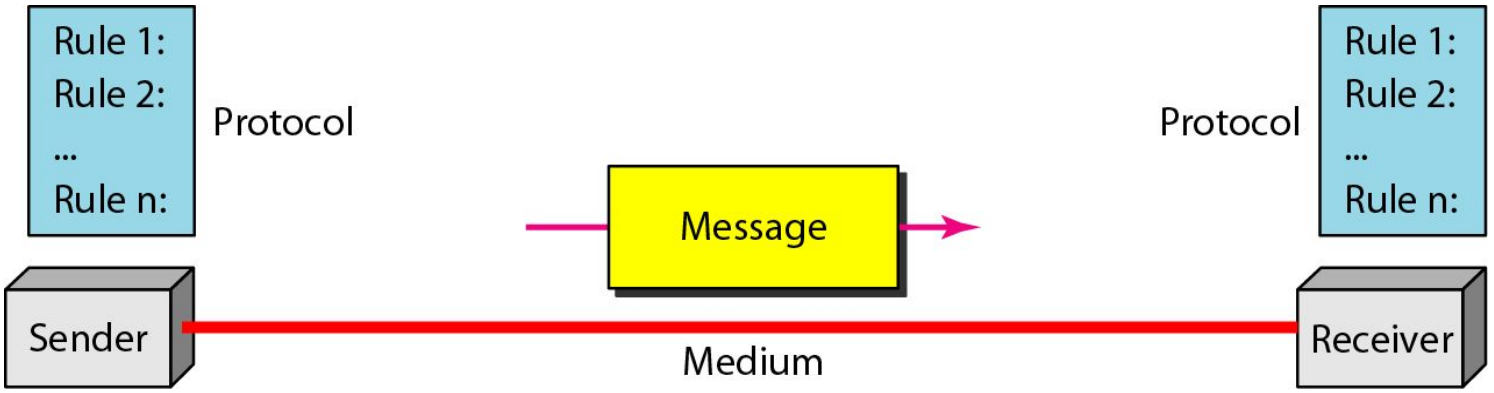
Data Communications

The term **Telecommunication** means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Five Characteristics of Data Communication (NETWORKS)

- Message:
  - text, number, images, audio, and video
- Sender and Receiver
  - devices that send/receive data message
  - Computer, workstation, telephone, TV, etc.
- Transmission medium
  - Physical path thru which the message travels
- Protocol
  - Set of rules governing data communications

Four Characteristics of Networks



CLASSIFICATION OF NETWORKS

Criteria

- Size
- Topology
- Architecture
- Administration

❖Computer Networks

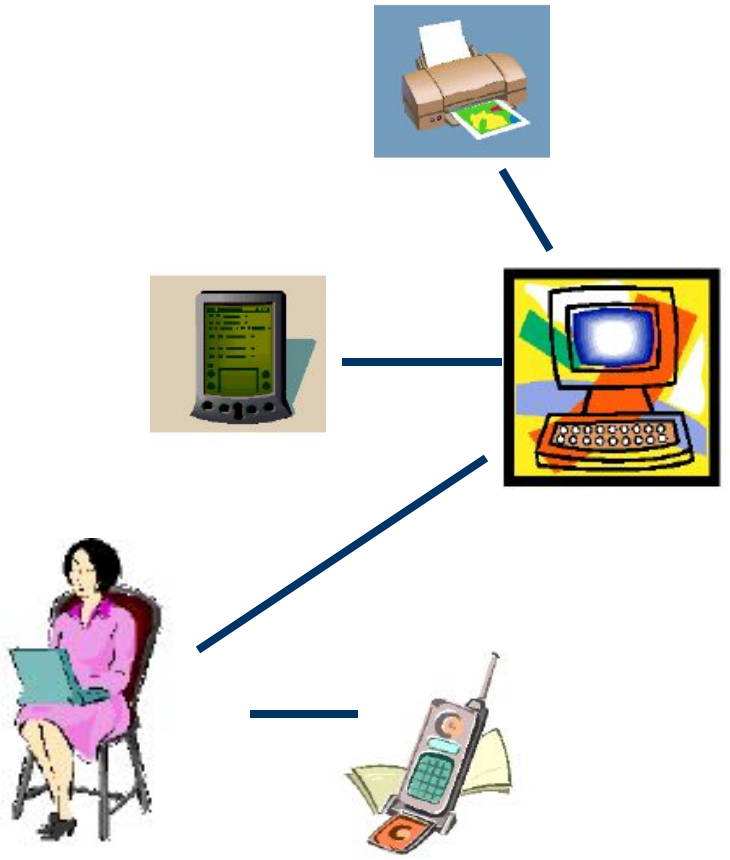
- *A computer network is a system for communicating between two or more computers and associated devices*
- A popular example of a computer network is the internet, which allows millions of users to share information
- Computer networks can be classified according to their **size:**
  - Personal area network (PAN)
  - Local area network (LAN)
  - Metropolitan area network (MAN)
  - Wide area network (WAN)



SIZE

Personal Area Network

- *A PAN is a network that is used for communicating among computers and computer devices (including telephones) in close proximity of around a few meters within a room*
- It can be used for communicating between the devices themselves, or for connecting to a larger network such as the internet
- PAN's can be wired or wireless
  - PAN's can be wired with a computer bus such as a universal serial bus: *USB* (a serial bus standard for connecting devices to a computer-many devices can be connected concurrently)
  - PAN's can also be wireless through the use of *bluetooth* (a radio standard designed for low power consumption for interconnecting computers and devices such as telephones, printers or keyboards to the computer) or *IrDA* (infrared data association) technologies



Local Area Network

- *A LAN is a network that is used for communicating among computer devices, usually within an office building, an organisation etc*
- LAN's enable the sharing of resources such as files or hardware devices that may be needed by multiple users
- Is limited in size, typically spanning a few hundred meters, and no more than a mile
- Is very fast, with speeds from 10 Mbps to 10 Gbps
- Requires very little wiring, typically a single cable connecting to each device
- Has lower cost compared to MAN's or WAN's

MAN – metropolitan area network

Networks connecting several Institutions in a City (Metro).

- Public
- High speed access
- Voice and data transmission
- Range about 50 miles

- Example – ISP, cable television, local telephone

Metropolitan-Area Network (MAN). This is a large communication infrastructure connecting organisations in large cities to facilitate E-Commerce and E-trade.

MAN Cont…

A Metropolitan Area Networks bridges a number of 'Local Area Networks' with a fiber-optical links which act as a backbone, and provides services similar to what Internet Service Provider (ISP) provide to Wide Area Networks and the Internet.

Major technologies used in MAN networks are 'Asynchronous Transfer Mode (ATM)', 'Fiber Distributed Data Interface (FDDI)' and 'Switched Multi-megabit Data Service (SMDS, a connectionless service)'.

WAN

Wide Area Network is a computer network that covers relatively larger geographical area such as a state, province or country. It provides a solution to companies or organizations operating from distant geographical locations who want to communicate with each other for sharing and managing central data or for general communication.

WAN is made up of two or more Local Area Networks (LANs) or Metropolitan Area Networks (MANs) that are interconnected with each other, thus users and computers in one location can communicate with users and computers in other locations.
In 'Wide Area Network', Computers are connected through public networks, such as the ***telephone systems, fiber-optic cables, and satellite links or leased lines.***

WAN Cont…..

The ‘Internet’ is the largest WAN in a world. WANs are mostly private and are built for a particular organization by ‘Internet Service Providers (ISPs)’ which connects the LAN of the organization to the internet.

WANs are frequently built using **expensive leased lines** where with each end of the leased line a router is connected to extend the network capability across sites.

ARCHITECTURE

Whata is peer-to-peer (p2p)?

1. „(…) a type of network in which each work station has equivalent capabilities and responsibilities”

source: webopedia.com

2. "individual computers may share hard drives, CD-ROM drives, and other storage devices with the other computers on the network." Contrast with this, in a client/server set-up, "most of the computers (clients) tend to share resources from one main computer (the server). "

source: p2p.inf.pl

3. Peer-to-peer is a communications model in which each party has the same capabilities and either party can initiate a communication session. Other models with which it might be contrasted include the client/server model and the master/slave model..."

source: searchNetworking .techtarget.com

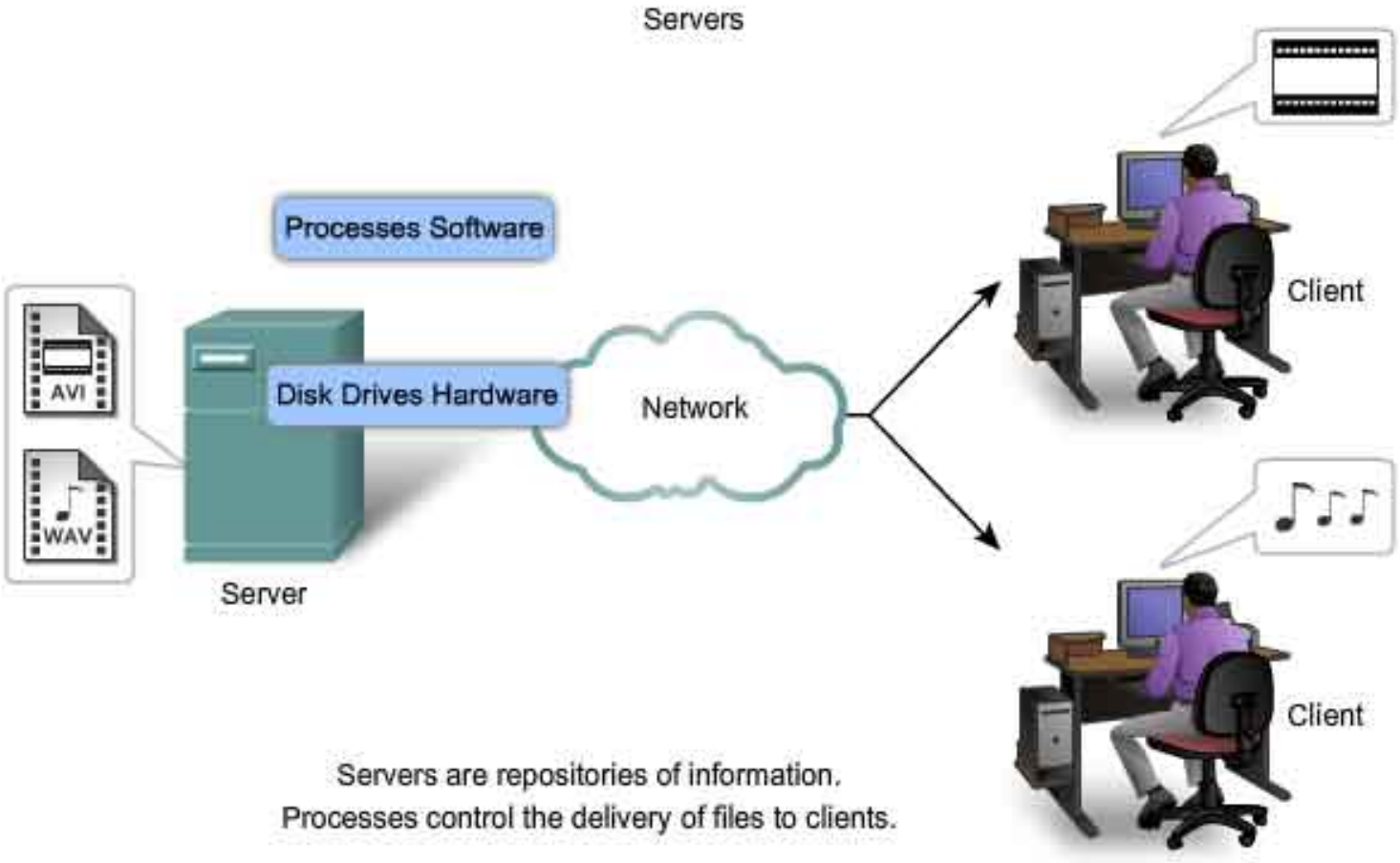
Advantages

- Easy to Install and Manage
- They are Cheap
- The more nodes that are part of the system, demand increases and total capacity of the system also increases. Where in client-server network architectures as more clients are added to the system, the system resources decreases.
- There is no single point of failure, due to robustness of the system.
- All clients provide to the system

Disadvantages

- Security is a major concern, not all shared files are from benign sources. Attackers may add malware to p2p files as an attempt to take control of other nodes in the network.
- Heavy bandwidth usage
- Potential legal/moral concerns

Client/Server Networking Model



Networking

- Computer networks have opened up an entire frontier in the world of computing called the **client/server model**

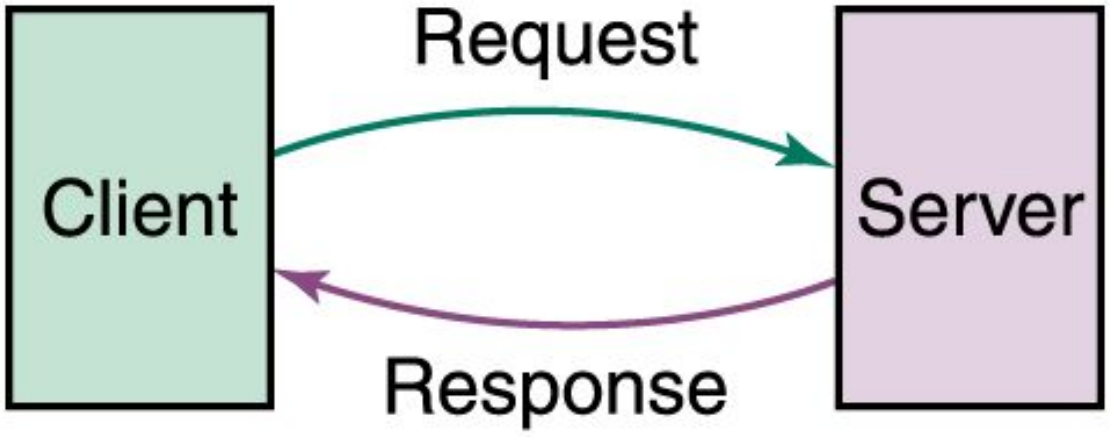


Figure 15.1 **Client/Server interaction**

Advantages/Disadvantages of Server/Client

- Faster and efficient
- Lower network traffic
- Improved processing distribution
- Thinner clients
- Greater processing transparency
- Increased network, hardware, and software transparency
- Improved security
- Decreased costs
- Increased scalability

- **Expensive to install and acquire**
- **Requires advanced Technical skills to operate**

Network Topologies

Objectives

- Define a Network Topology ?
- Describe the basic and hybrid LAN physical topologies, and their uses, advantages and disadvantages
- Describe the backbone structures that form the foundation for most LANs

Network topology

- **TWO TYPES OF TOPOLOGIES**:

Physical and Logical

- Physical topology of a network refers to the layout of cables, computers and other peripherals
  - network cables coming out of every computer that is part of the network, then those cables plug into a hub or switch
- Logical topology is the method used to pass the information between the computers
  - the network works with all the computers talking (think of the computers generating traffic and packets of data going everywhere on the network)
  - computers will be talking to each other and the direction of the traffic is controlled by the various protocols (like Ethernet)

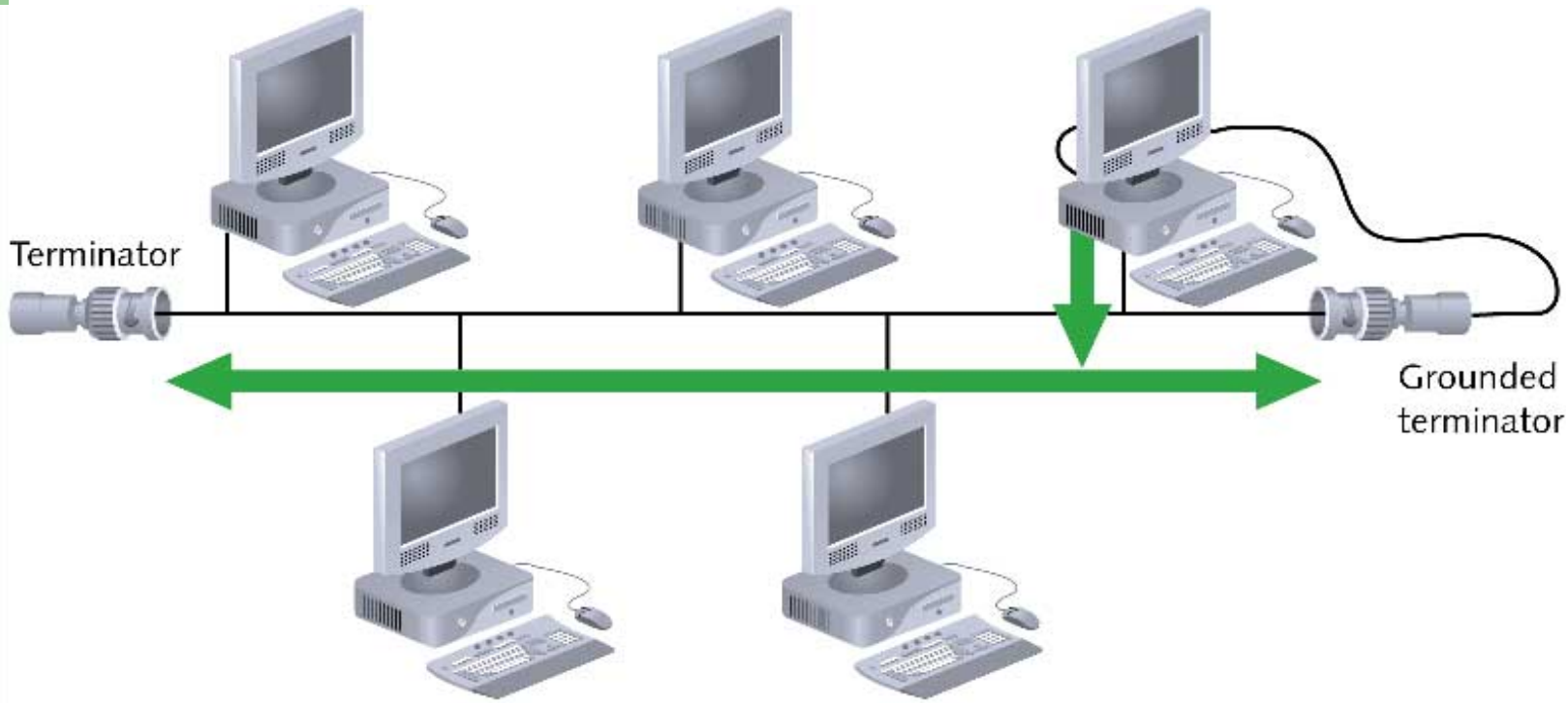
Simple Physical Topologies

- Physical topology: physical layout of nodes on a network
- Three fundamental shapes:
  - Bus
  - Ring
  - Star
- May create hybrid topologies
- Topology integral to type of network, cabling infrastructure, and transmission media used

Bus

- Single cable connects all network nodes without intervening connectivity devices
- Devices share responsibility for getting data from one point to another
- Terminators stop signals after reaching end of wire
  - Prevent signal bounce
- Inexpensive, not very scalable
- Difficult to troubleshoot, not fault-tolerant

Bus (continued)



Advantages of Bus Topology

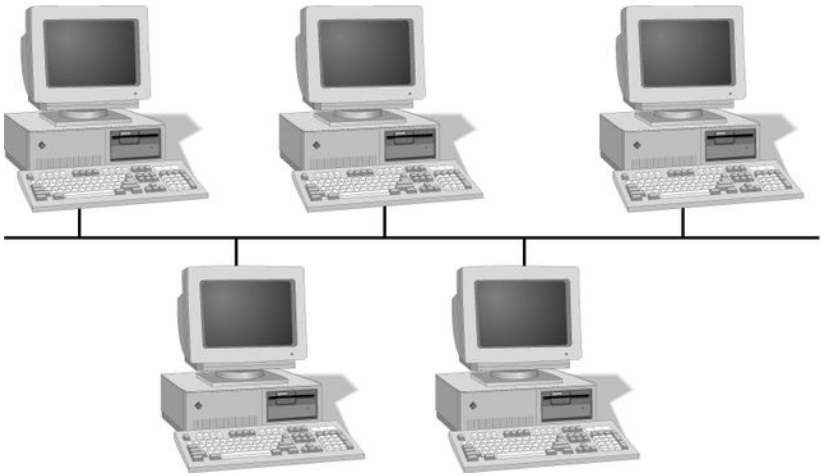
- Works well for small networks
- Relatively inexpensive to implement
- Easy to add to it

Disadvantages of Bus Topology

- Management costs can be high
- Potential for congestion with network traffic

Simple Physical Topologies

- Physical topology
  - Physical layout of a network
- A **Bus topology** consists of a single cable—called a **bus**— connecting all nodes on a network without intervening connectivity devices



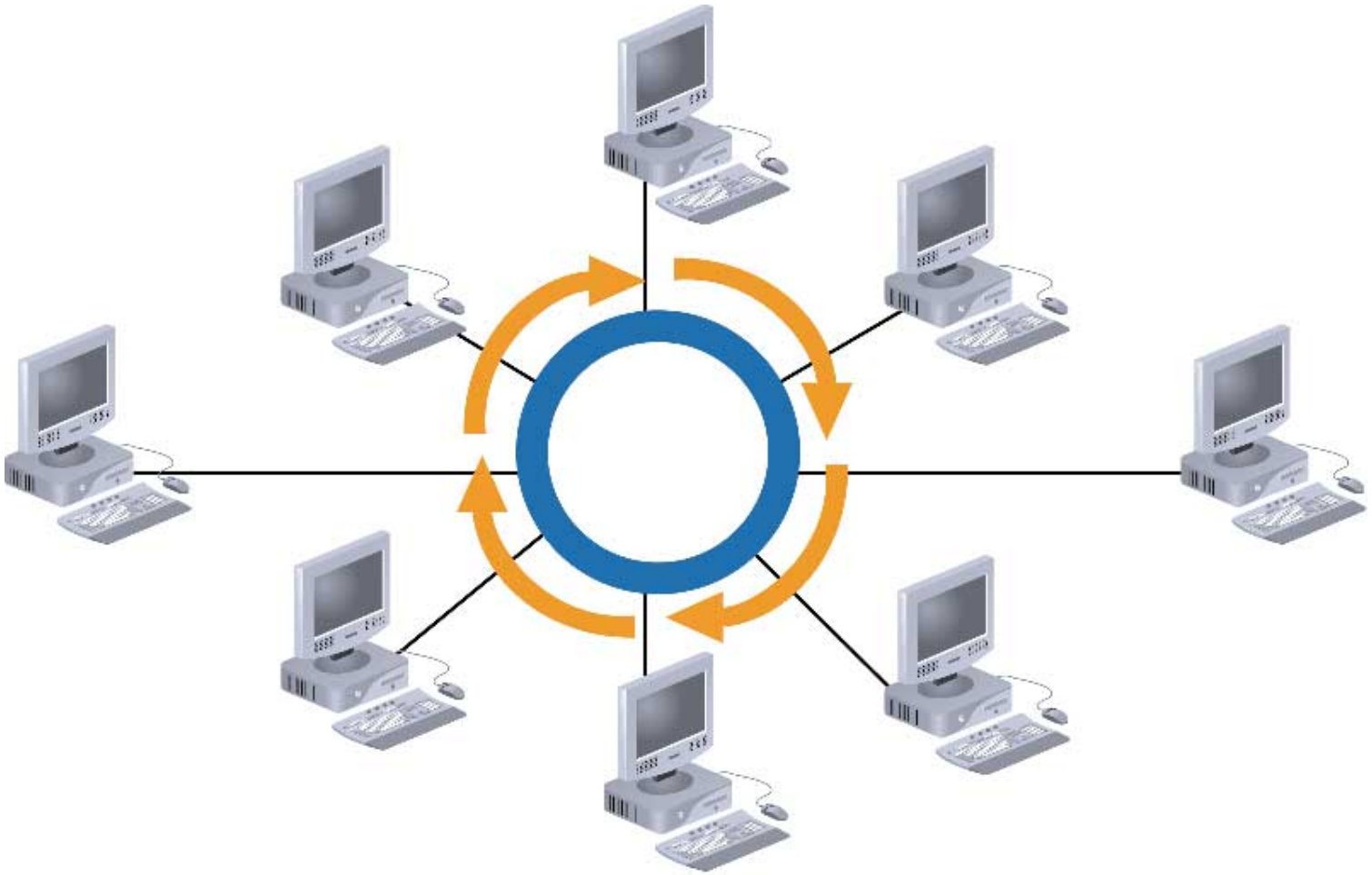
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Disadvantages of Bus Topology

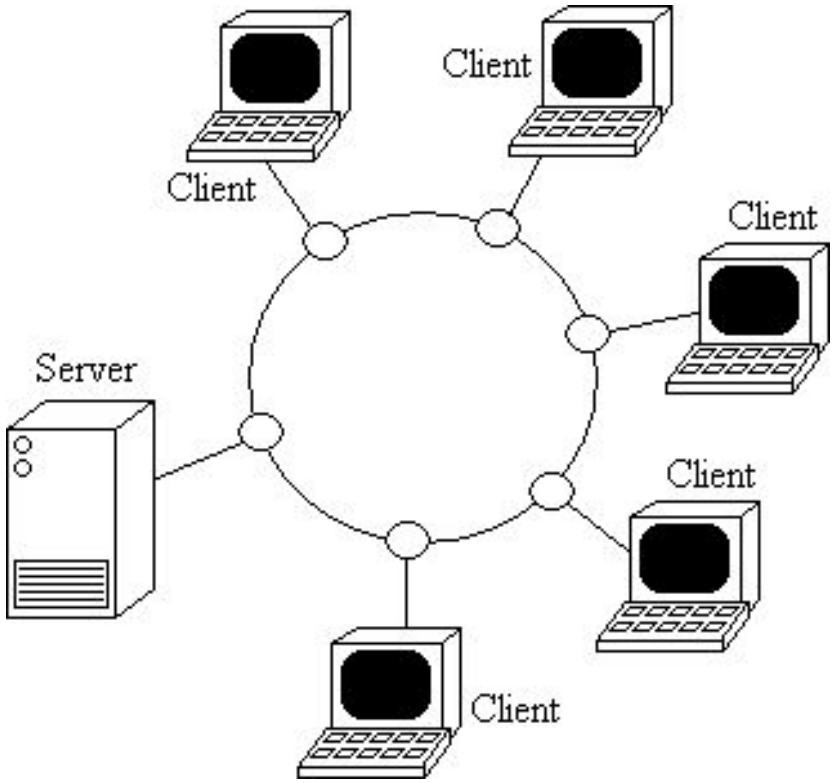
- Management costs can be high
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Ring



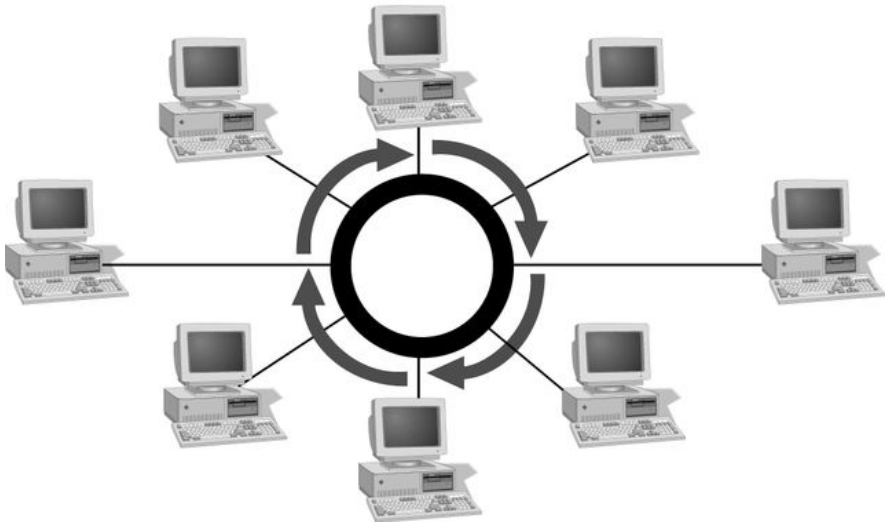
Ring

- All clients have equal access to data.
- This gives even performance under heavy demand conditions.
- Difficult to add another host to the ring.



Simple Physical Topologies

- Ring topology
  - Each node is connected to the two nearest nodes so the entire network forms a circle
  - One method for passing data on ring networks is **token passing**
- Active topology
  - Each workstation transmits data



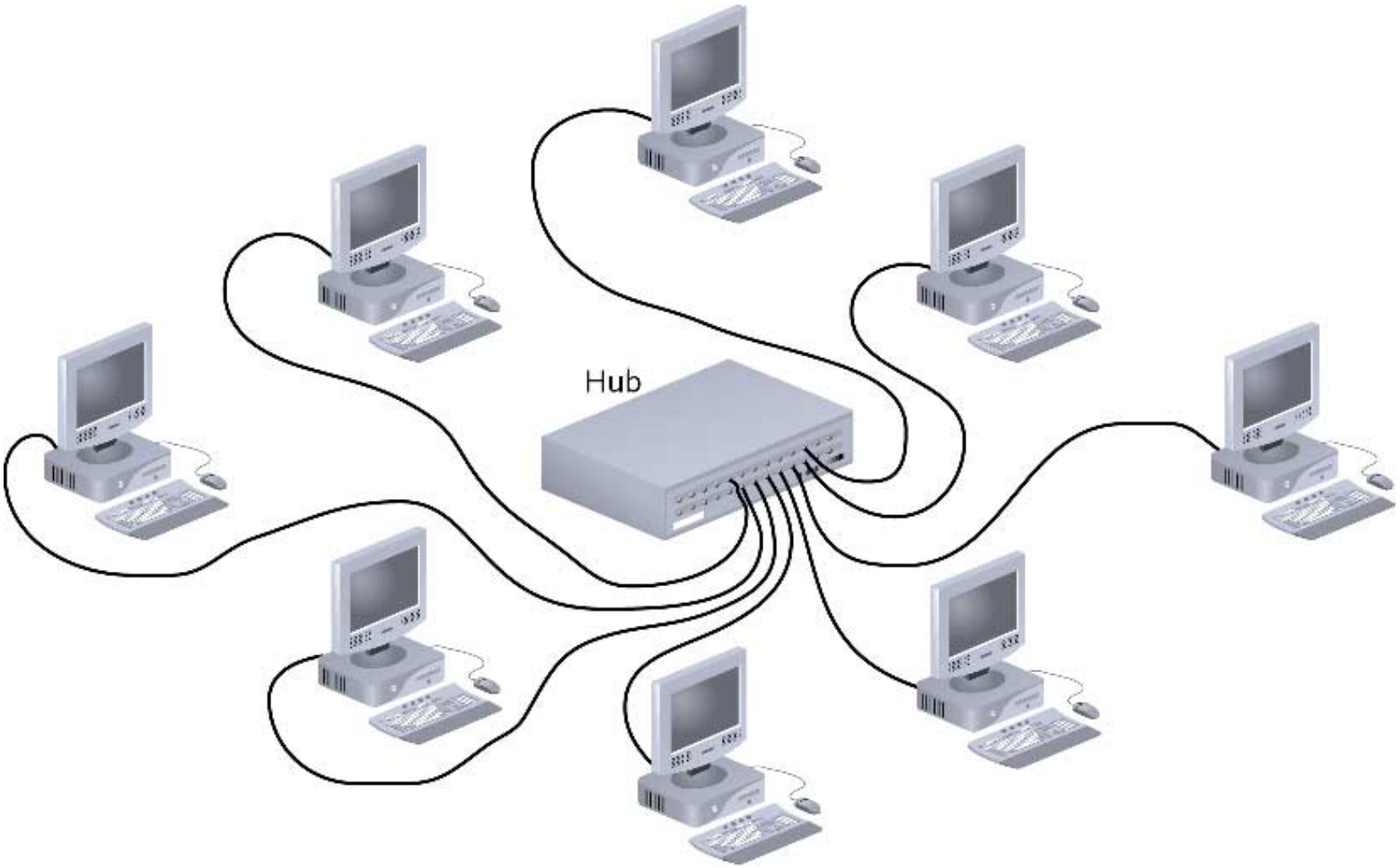
Advantages of Ring Topology

- Easier to manage; easier to locate a defective node or cable problem
- Well-suited for transmitting signals over long distances on a LAN
- Handles high-volume network traffic
- Enables reliable communication

Disadvantages of Ring Topology

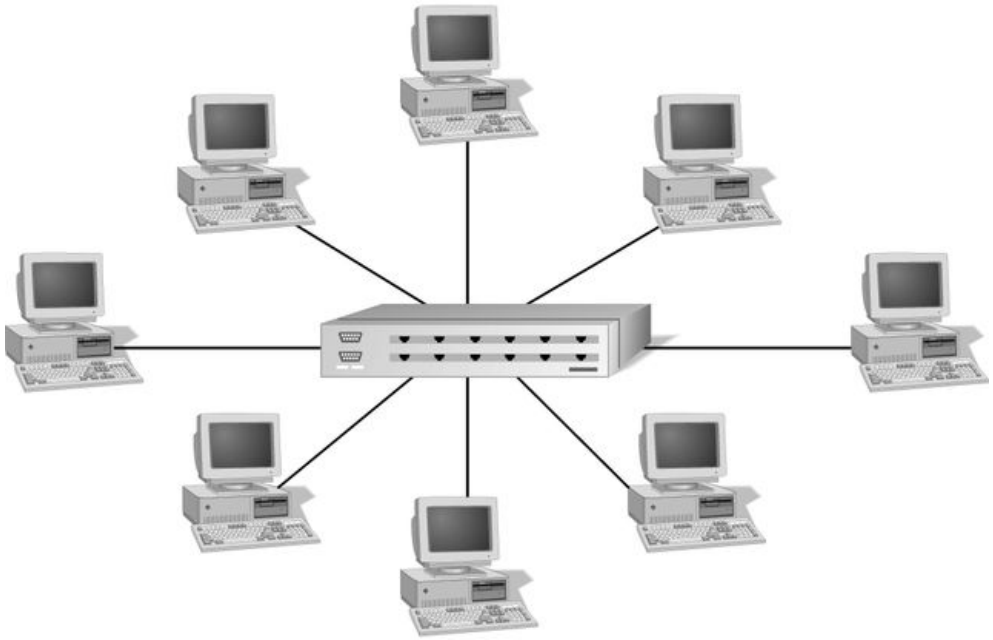
- Expensive
- Requires more cable and network equipment at the start
- Not used as widely as bus topology
  - Fewer equipment options
  - Fewer options for expansion to high-speed communication

Star

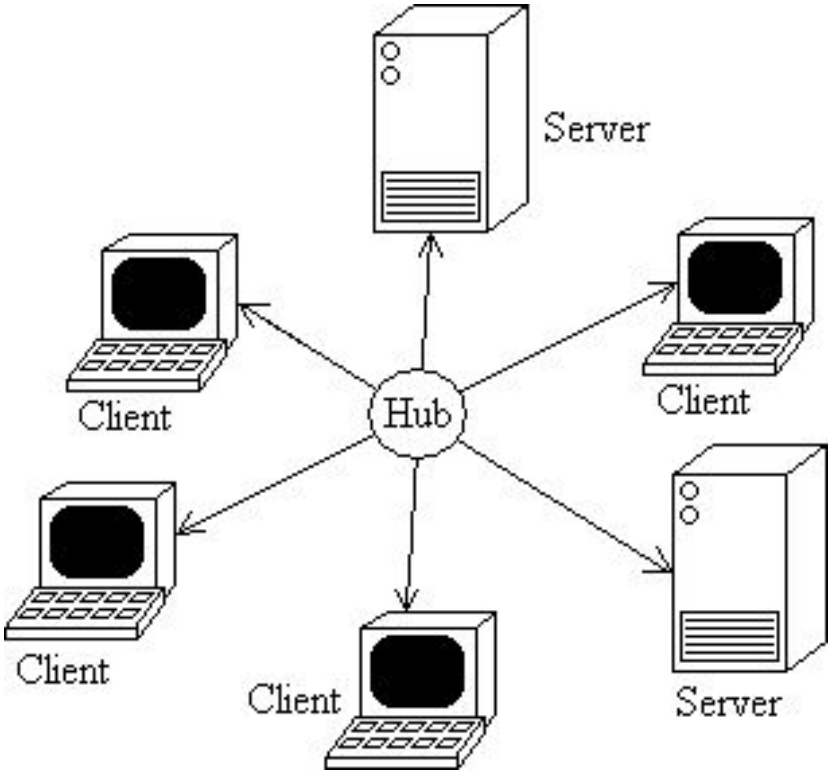


Simple Physical Topologies

- Star topology
  - Every node on the network is connected through a central device



Star



- Reduces risk of failure.
- Expanding the network is simple.
- Managing the network is centralized.
- If the hub breaks down, the system fails.

Star (continued)

- Any single cable connects only two devices
  - Cabling problems affect two nodes at most
- Requires more cabling than ring or bus networks
  - More fault-tolerant
- Easily moved, isolated, or interconnected with other networks
  - Scalable
- Supports max of 1024 addressable nodes on logical network

Advantages of Star Topology

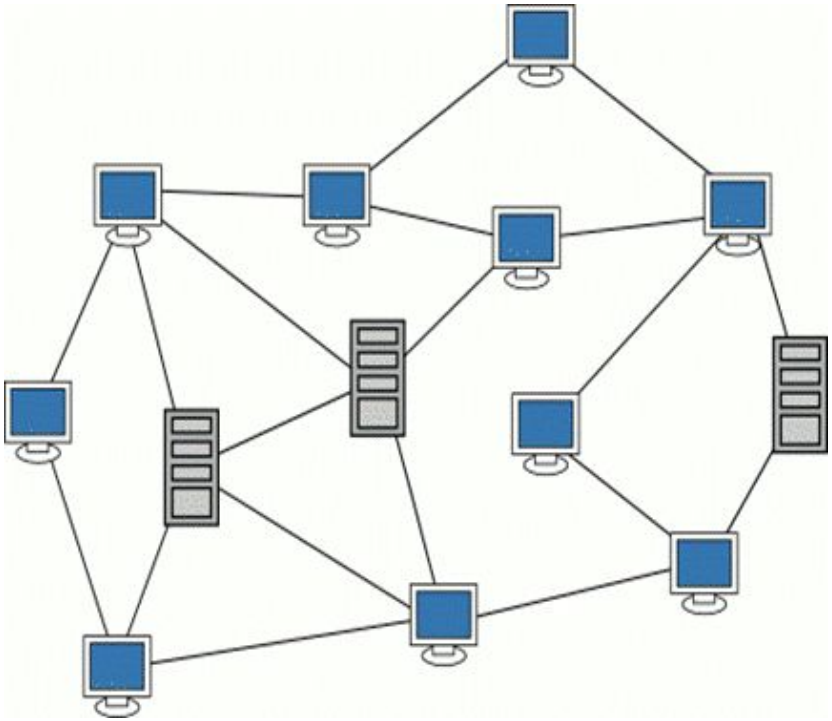
- Good option for modern networks
- Low startup costs
- Easy to manage
- Offers opportunities for expansion
- Most popular topology in use; wide variety of equipment available

Disadvantages of Star Topology

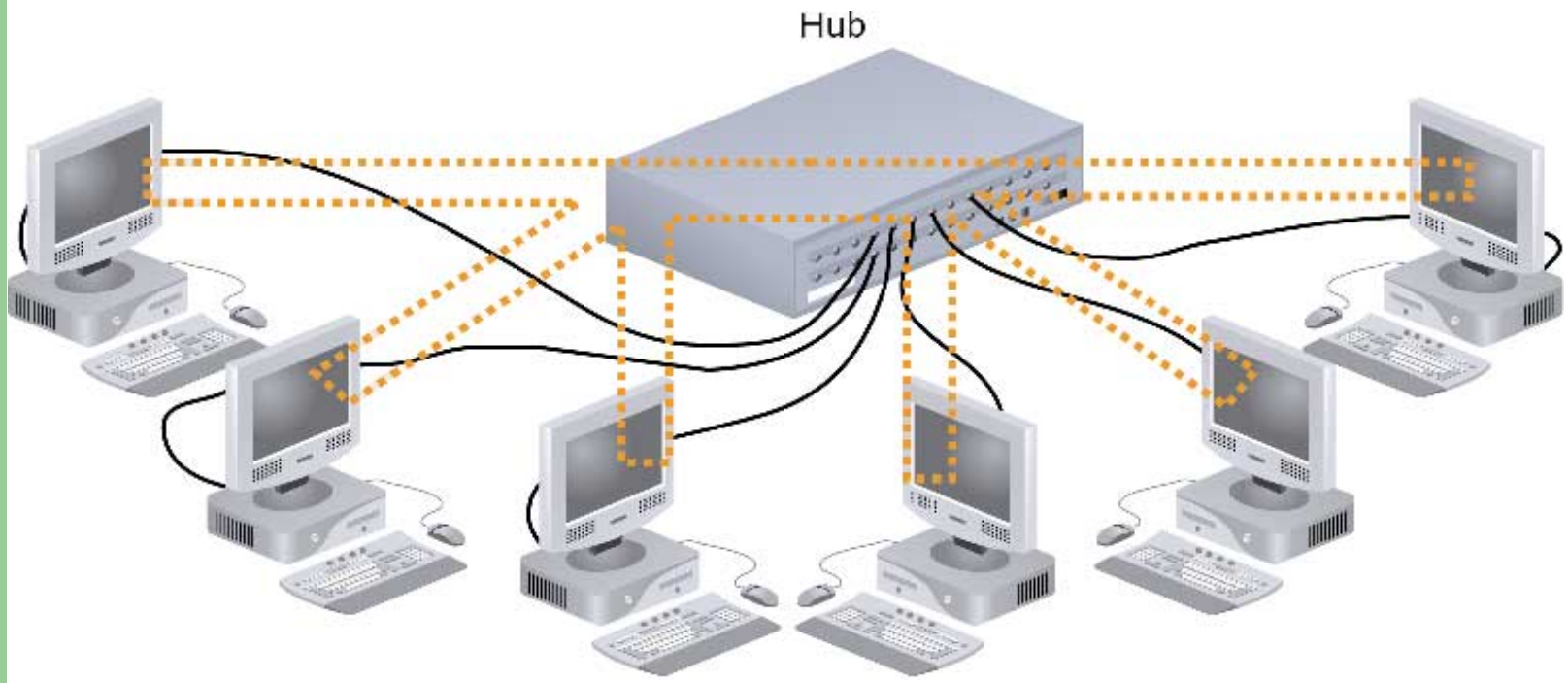
- Hub is a single point of failure
- Requires more cable than the bus

Mesh

- This redundancy allows the system to continue if some connections fail.
- However, multiple connections are expensive.

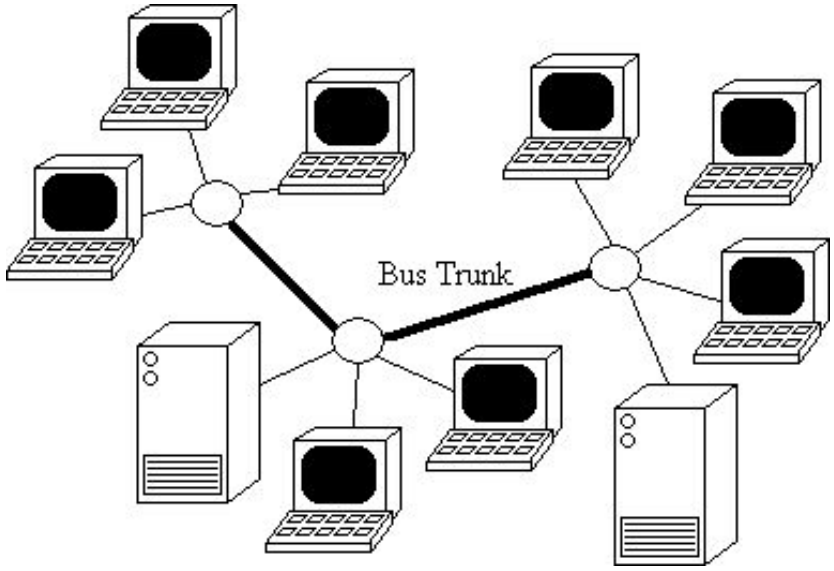


Hybrid Physical Topologies: Star-Wired Ring

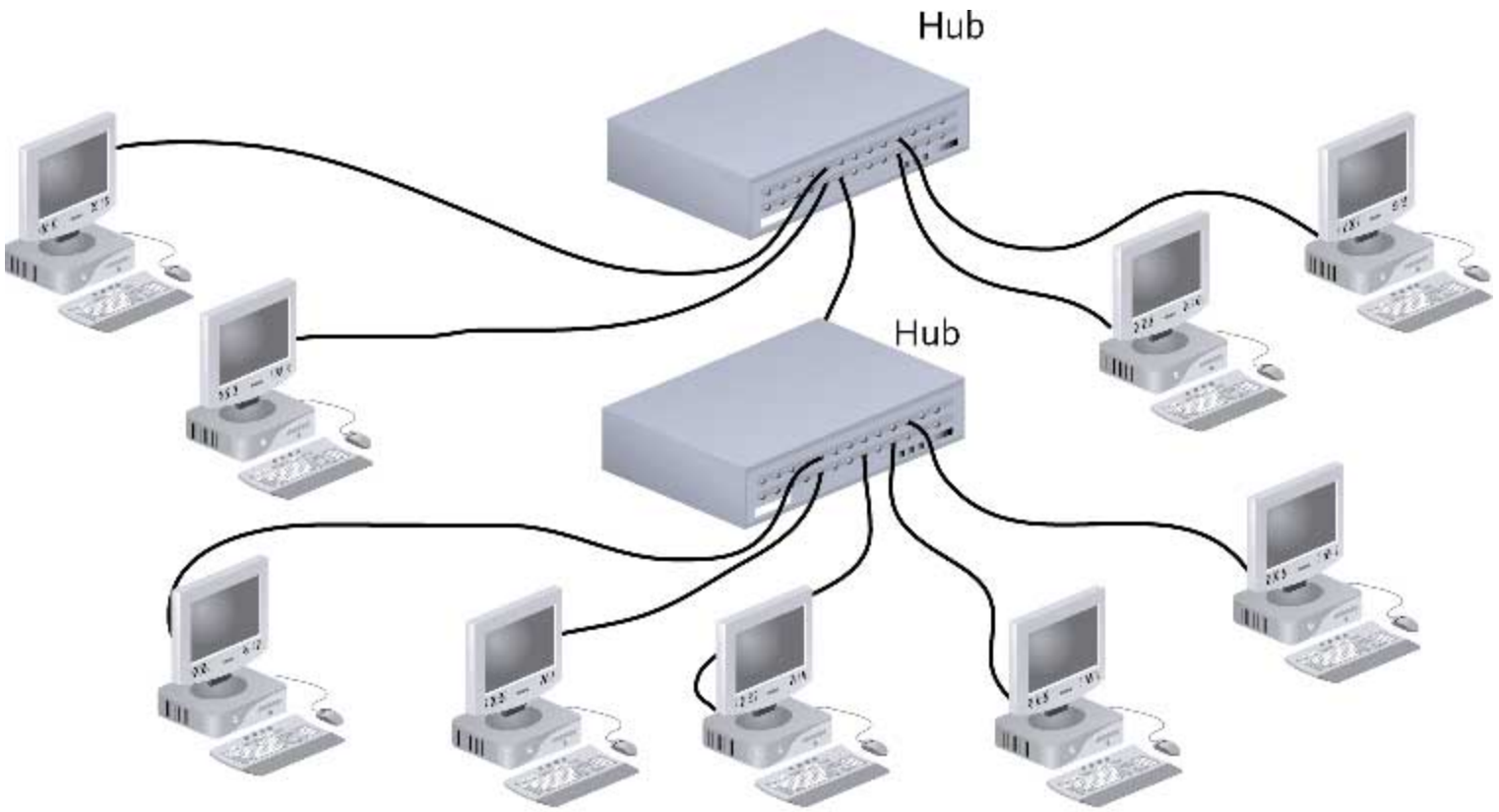


Hybrid – Star and Bus

- Network expansion is simple.
- If one client fails, the entire network does not fail.
- If one hub fails, all connections to that hub fail, although other hubs continue to function.



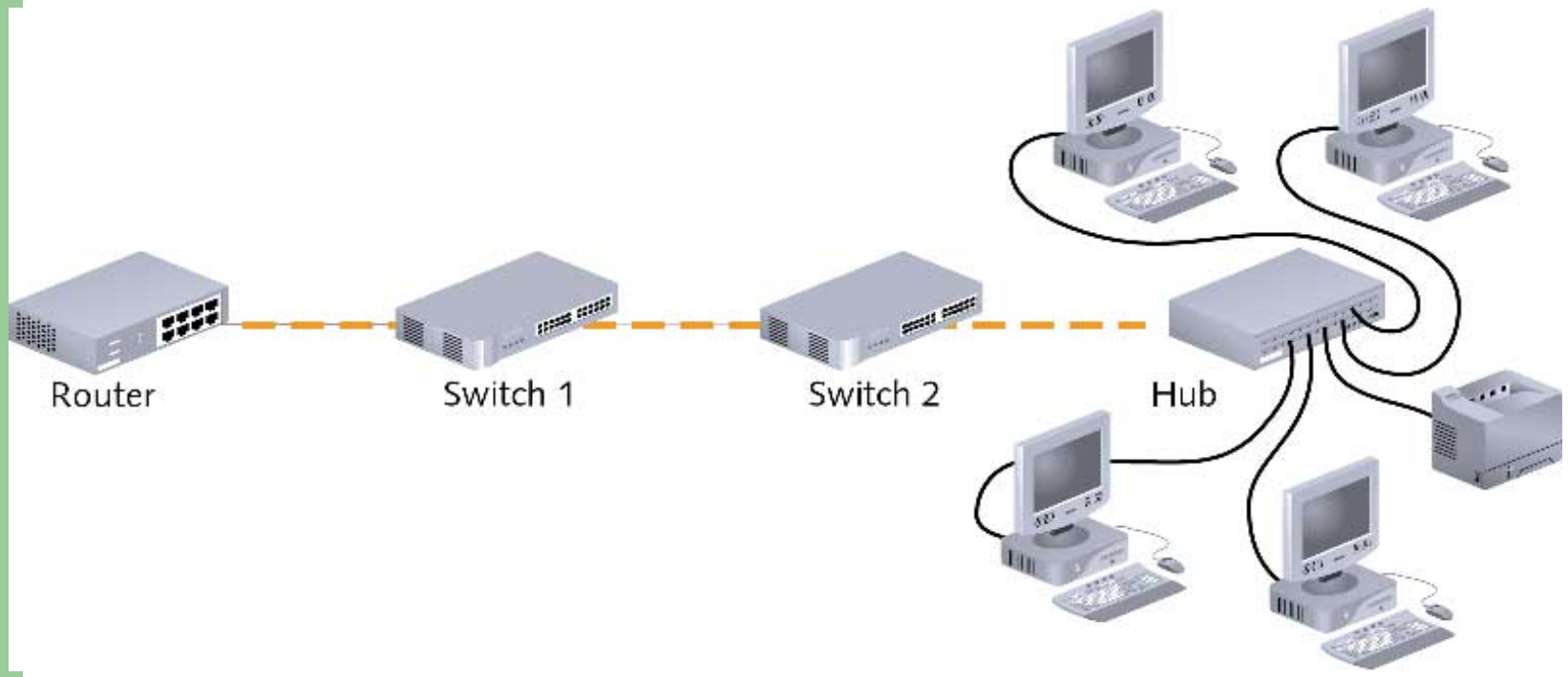
Star-Wired Bus



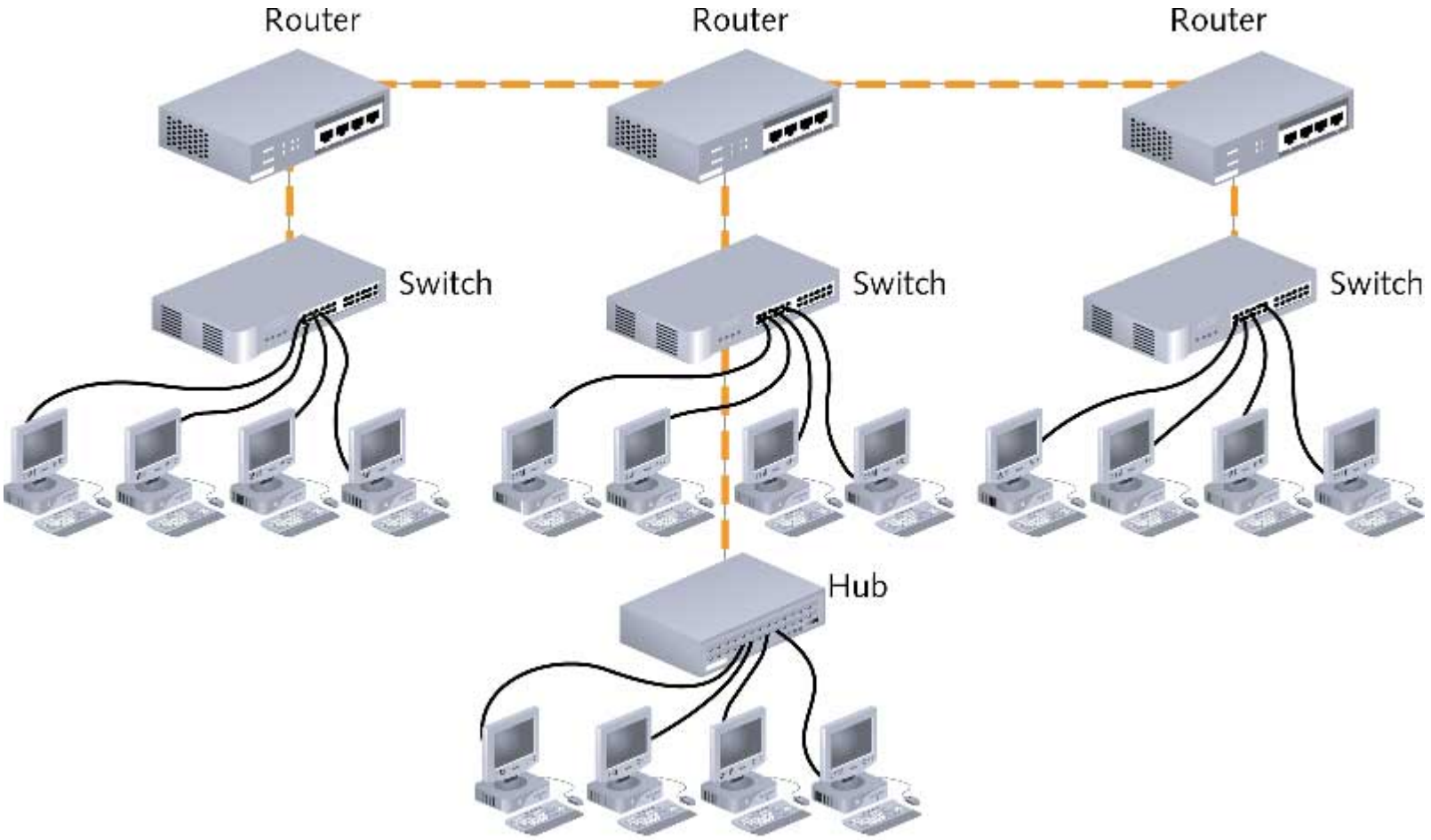
Backbone Networks: Serial Backbone

- Daisy chain: linked series of devices
  - Hubs and switches often connected in daisy chain to extend a network
- Hubs, gateways, routers, switches, and bridges can form part of backbone
- Extent to which hubs can be connected is limited

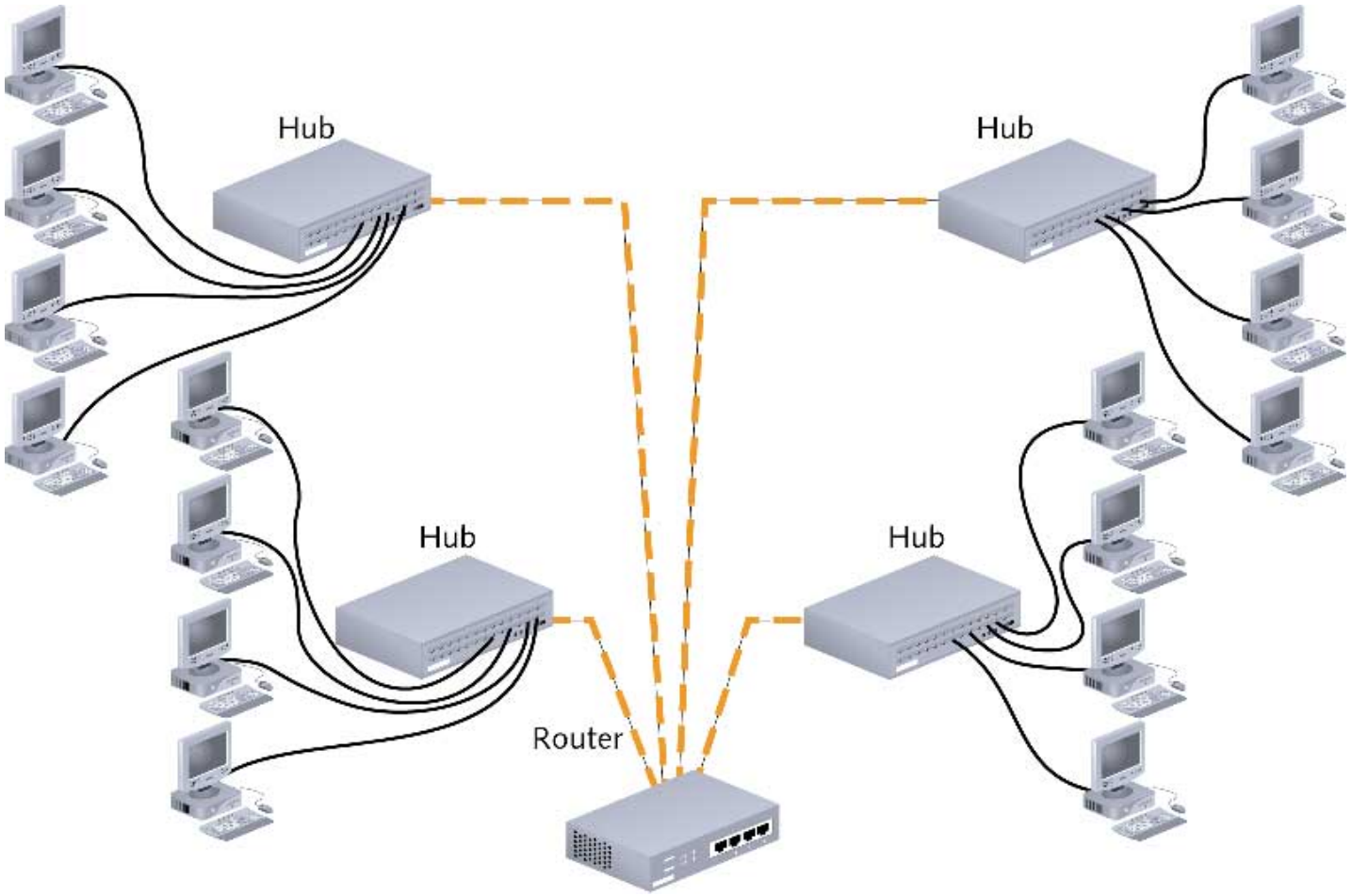
Backbone Networks: Serial Backbone (continued)



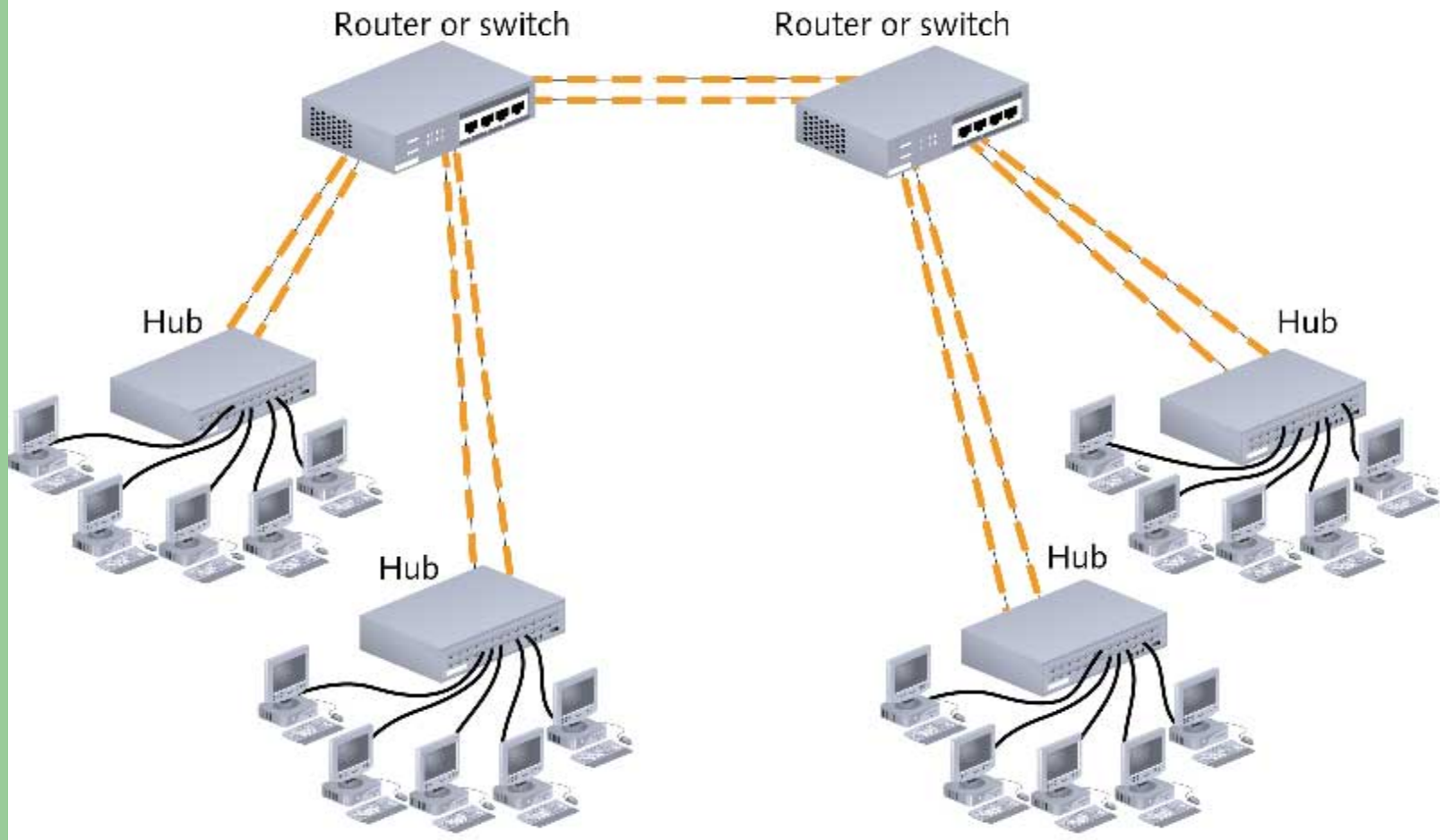
Distributed Backbone



Collapsed Backbone



Parallel Backbone



Logical Topologies

- Logical topology: how data is transmitted between nodes
  - May not match physical topology
- Bus logical topology: signals travel from one network device to all other devices on network
  - Required by bus, star, star-wired physical topologies
- Ring logical topology: signals follow circular path between sender and receiver
  - Required by ring, star-wired ring topologies

Network Devices

Functions of network devices

- **Separating (connecting)** networks or **expanding** network
  - e.g. repeaters, hubs, bridges, routers, brouters, switches, gateways
- **Remote access**
  - e.g. 56K Modems and ADSL modems

A. Expanding Network

- Networks cannot be made larger by simply adding new computers and more cables
  - Less efficient !!

- Can install components to
  - segment (divide) large LAN to form smaller LANs
  - connect LANs
- Required components
  - Repeaters, bridges, routers, brouters, switches or gateways

RJ-45 CONNECTOR



RJ-45 stands for **R**egistered **J**ack- 45. It is an eight wire connector which is commonly used to connect computers on local area networks i.e. LAN.

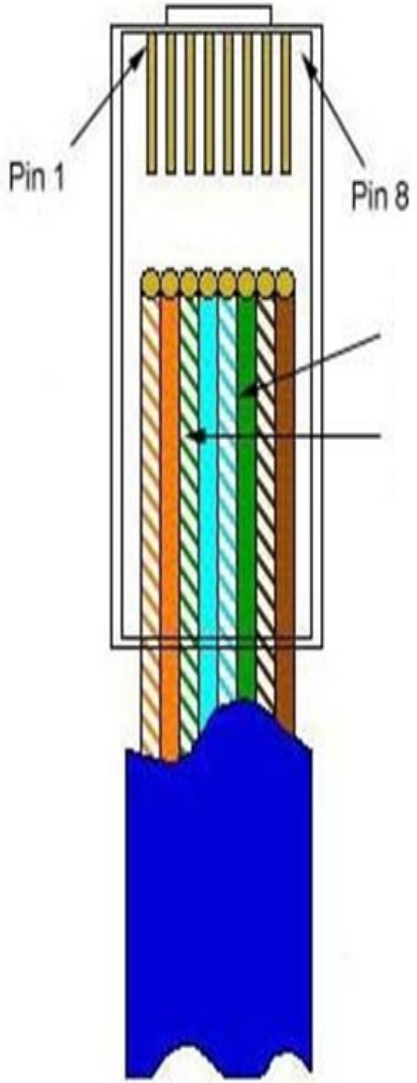


RJ-45 Connector is much wider than RJ-11 as it houses 8 wires instead of four





Pins 1 & 2 are the Transmit Pins
Pins 3 & 6 are the Receive Pins
Pins 4, 5, 7, and 8 are not used



Notice that the Green pair has to be seperated before crimping on the RJ45 plug. If this is not done, the Recieve Data will be split between 2 pairs and performance will degrade!

ETHERNET TECHNOLOGY

Ethernet is the dominant cabling and low level data delivery technology used in Local Area Networks (LAN's).

It was developed by Xerox corp. along with DEC and Intel.

Features:

1. Ethernet transmits data up to **ten million bits per second.**
2. Ethernet supports networks built with twisted pair, thin and thick coaxial and fiber optic cabling.
3. Each device on an Ethernet network operates independently and equally, precluding the need for a central controlling device.
4. Ethernet supports a wide array of data types, including TCP/IP, ,apple talk, and IPX.
5. To prevent the loss of data, when two or more devices attempt to send packets at the same time, Ethernet detects collisions.

ETHERNET CARD

(NETWORK INTERFACE CARD)

A network interface card is a piece of computer hardware designed to allow computers to communicate over a computer network. Converts the data into BITS and transmits the same onto the network (which may be wired or wireless). The biggest job is putting the data onto the network for transmission.

The computers that are part of Ethernet have to install a special card called Ethernet Card.

An Ethernet supports three types of connections:

1. RJ-45 connection, if it is designed for twisted pair
2. BNC Connector, if it is designed for coaxial cable
3. AUI connector


PHONE
LINE
FM-56PCI-HSFi V:RW-01 552-0030-0100
MADE IN TAIWAN
Atech
ATS-354
CONEXANT
HSFi
CX11252-11
E136147.1
0125 MEXICO
LH1546

FUNCTIONING OF A HUB:

HUB

A Hub is a hardware device used to connect several computers together. A hub contains multiple ports. It is sometimes called a Concentrator.

Hubs forward any data packets – e.g. e-mail, word documents, spreadsheets, graphics- they receive over ***one port to all of the remaining ports***. All users connected to a single hub are in the same segment.

All computers/devices are compete for attention because the hub takes the data that comes into a port and sends it out all the other ports in the hub.

Resides on Layer 1 of the OSI model

PASSIVE HUBS

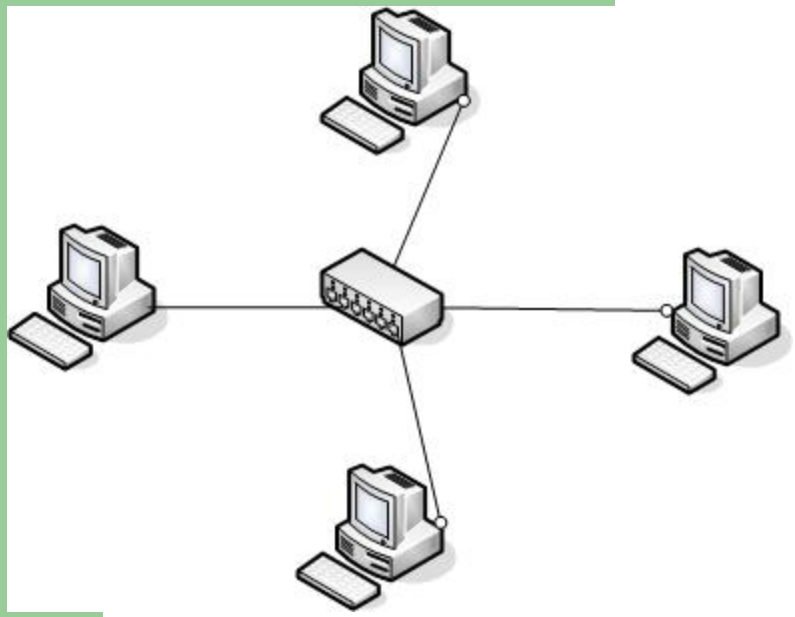
Allows the signal to pass from one computer to another without any change. It does not provide any processing or regeneration of signals.

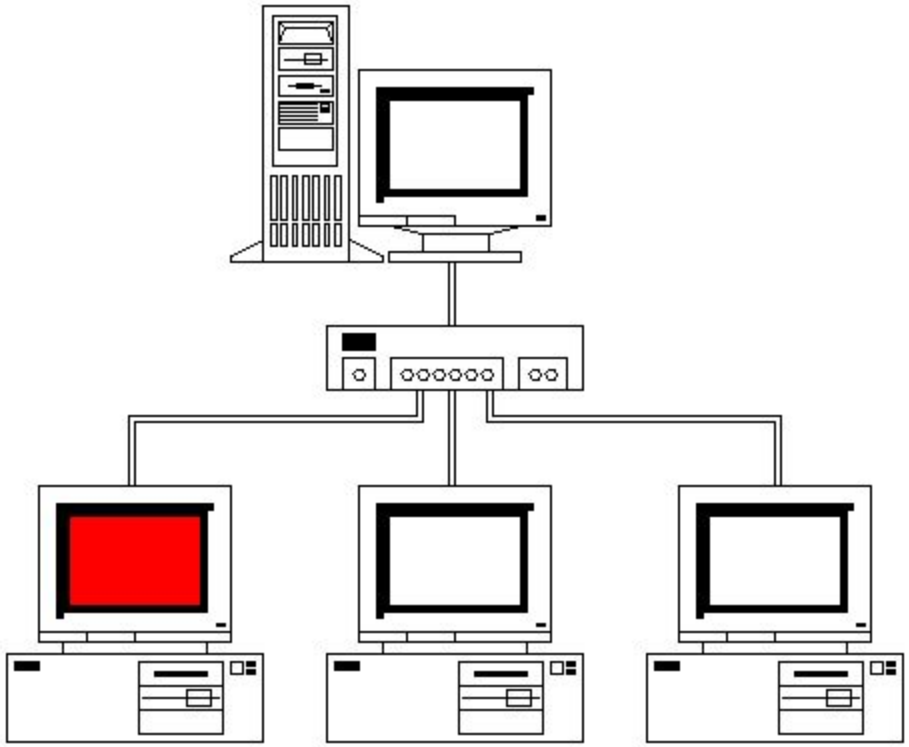
ACTIVE HUBS

Electrically **amplifies or strengthens** the signal as it moves from one connected device to the other.

SMART HUBS

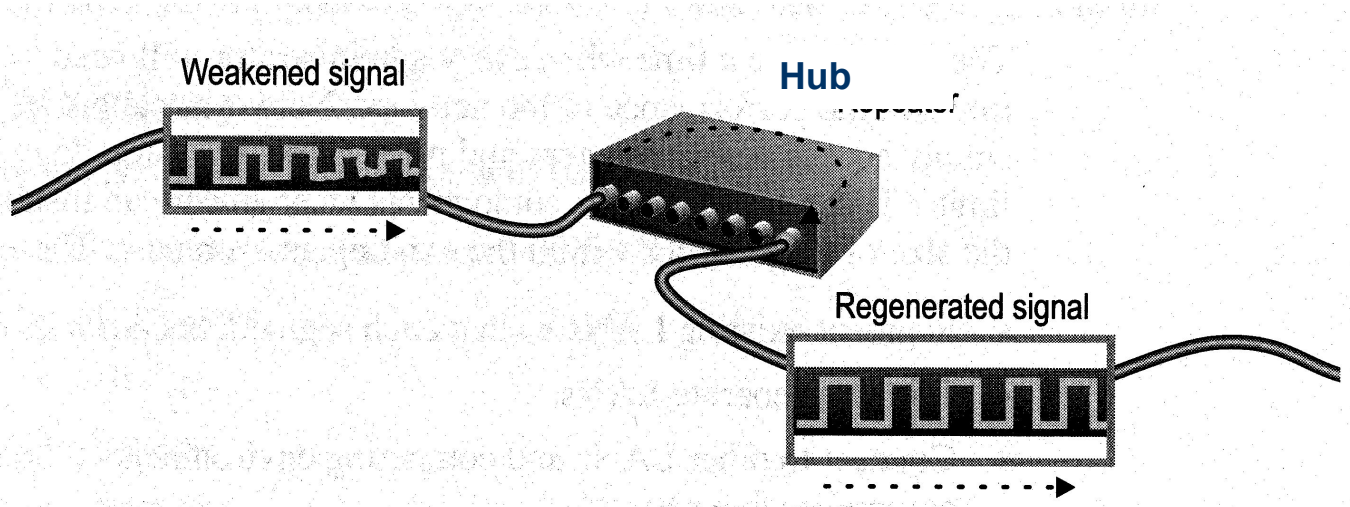
Electrically **amplifies or strengthens** the signal as it moves from one connected device to the other. In addition, they have installed Software that helps in NETWORK MANAGEMENT through SNMP Protocol.





REPEATERS

- Repeaters work at the OSI physical layer to regenerate the network's signal and resend them to other segments. It regenerates data and broadcasts them to all ports



Repeater

- Used to boost the signal between two cable segments or wireless access points.
- Can not connect different network architecture.
- Does not simply amplify the signal, it regenerates the packets and retimes them.
- Resides on Layer 1 of the OSI model.

OSI Model Layers

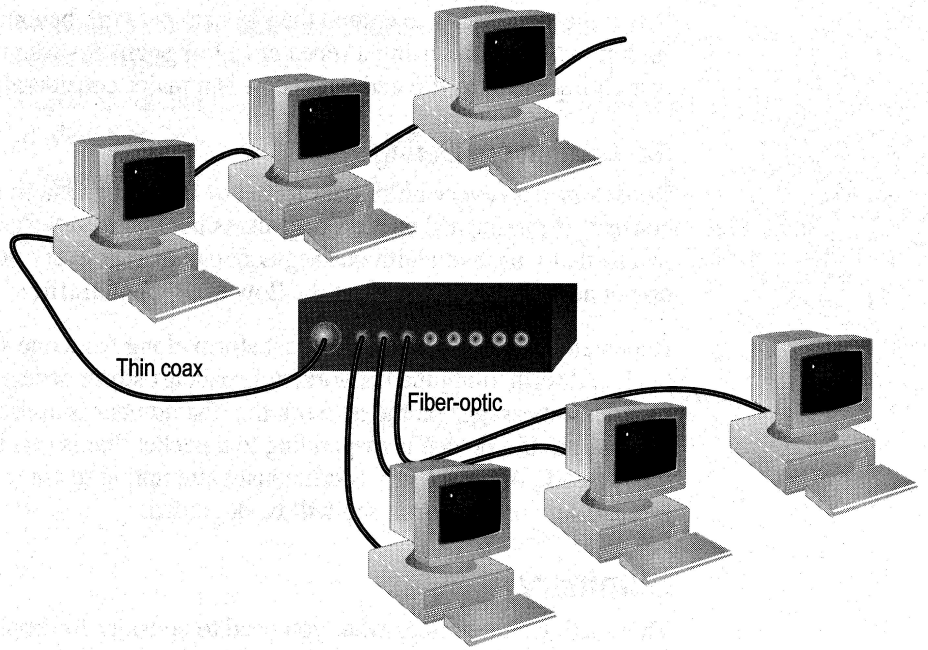
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical



Limitations and Features

- Cannot link unlike segments
- Cannot join segments with different access methods (e.g. CSMA/CD and token passing)
- Do not isolate and filter packets

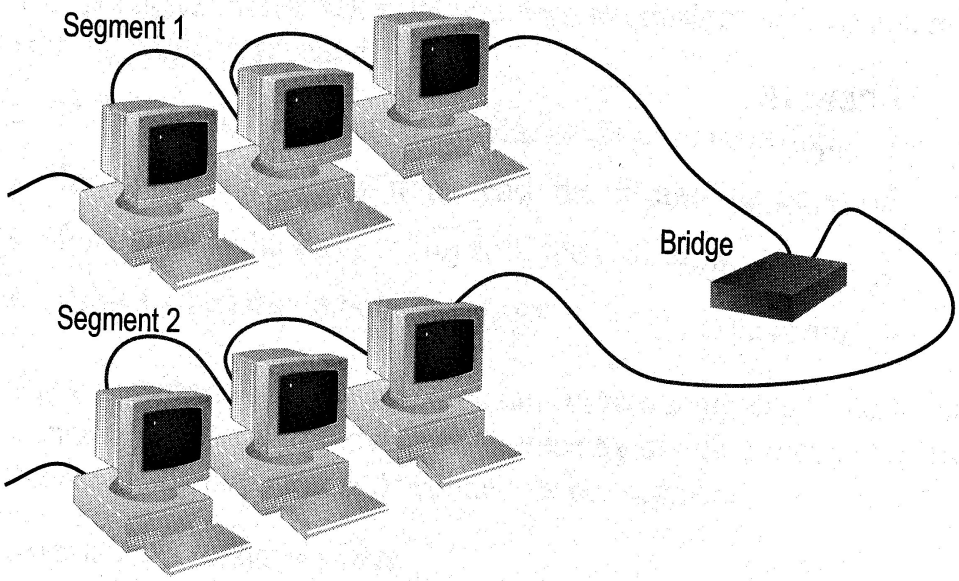
- Can connect different types of media
- The most economic way of expanding networks



b. Bridges

- Has one input and one output
- Used to isolate network traffic and computers
- Has the intelligent to examine incoming packet source and destination addresses

- But cannot interpret higher-level information
- Hence cannot filter packet according to its protocol



Bridge

- Connects two LANs and forwards or filters data packets between them.
- Creates an extended network in which any two workstations on the linked LANs can share data.
- Transparent to protocols and to higher level devices like routers.
- Forward data depending on the Hardware (MAC) address, not the Network address (IP).
- Resides on Layer 2 of the OSI model.

OSI Model Layers

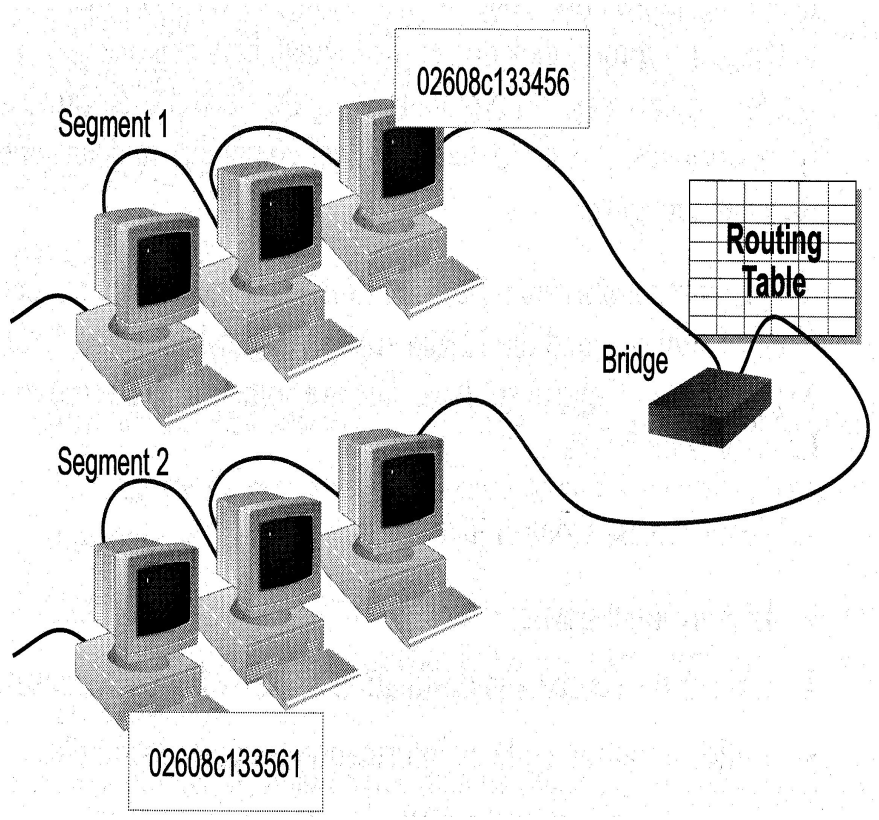
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical



How Bridges Work

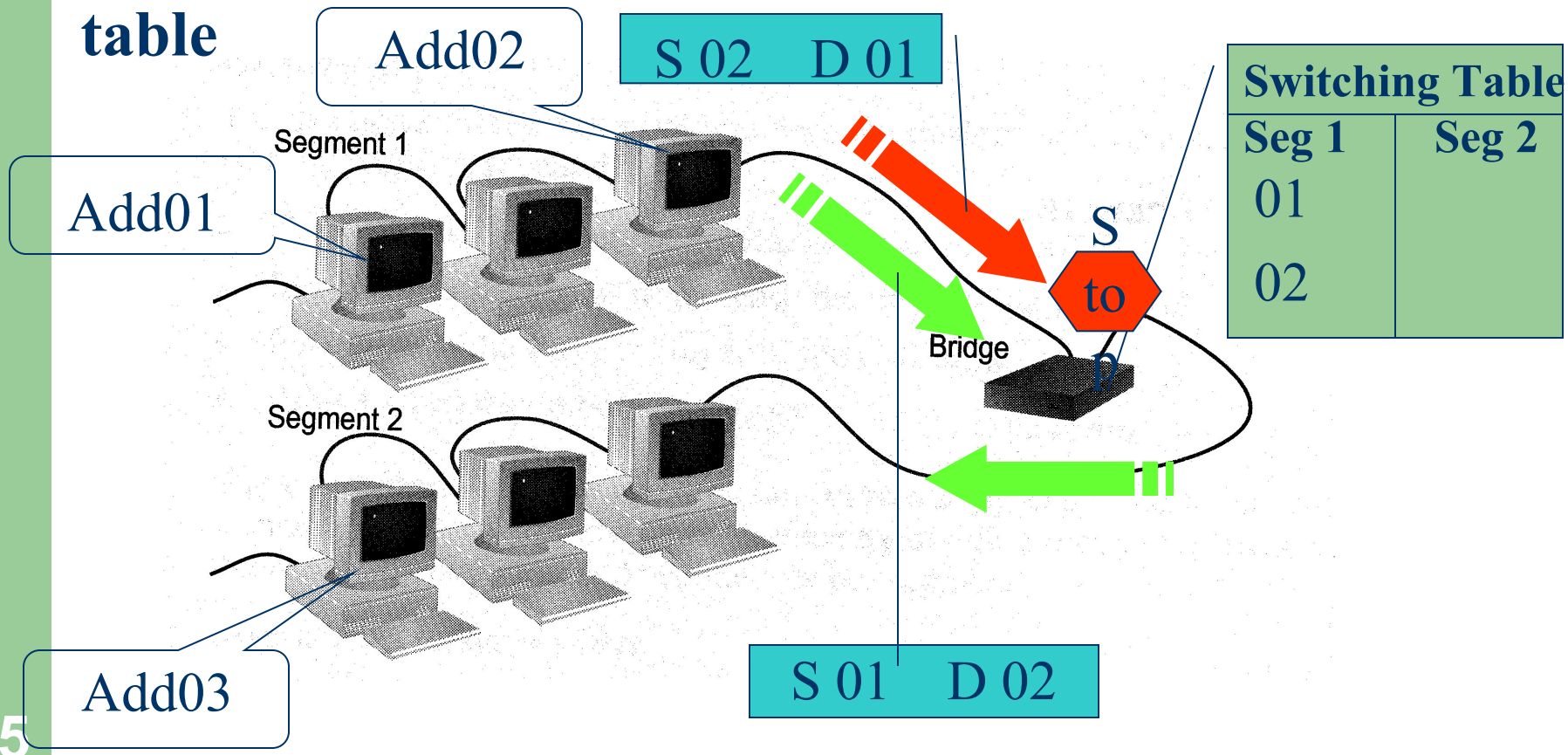
- Bridges work at the Media Access Control Sub-layer of the OSI model

- Routing table is built to record the segment no. of address
- If destination address is in the same segment as the source address, stop transmit
  Otherwise, forward to the other segment



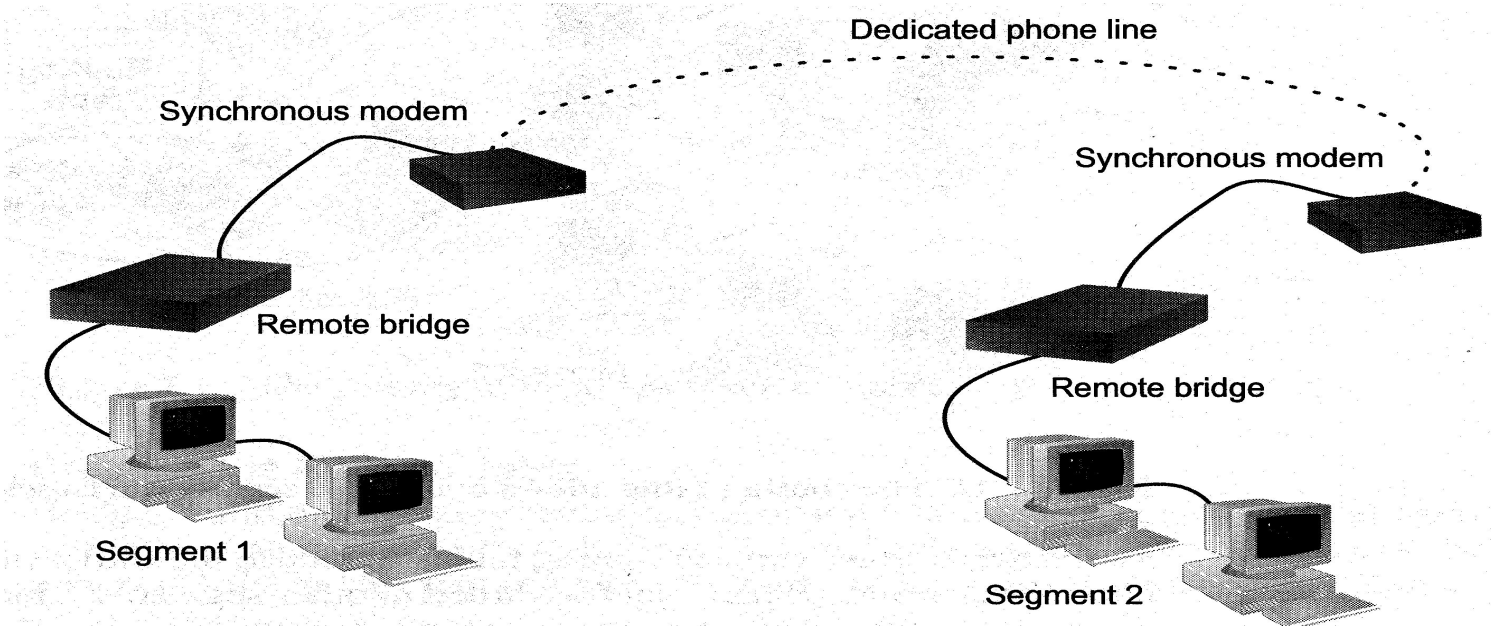
Creating a Switching Table

- Based on the addresses of the sending computers
- New addresses are added if they are not in the table



Remote Bridges

- Bridges are often used in large networks that have widely dispersed segments. Remote bridges can be used to connect remote segments via data-grade telephone line



Differences Between Bridges and Repeaters

	Repeaters	*Bridges*
OSI layer	Physical layer	Data link layer
Data regeneration	Regenerate data at the signal level	Regenerate data at the packet level
Reduce network traffic	No	Yes

SWITCHES

- Switches operate at the Data Link layer (layer 2) of the OSI model
- Can interpret address information
- Switches resemble bridges and can be considered as multiport bridges

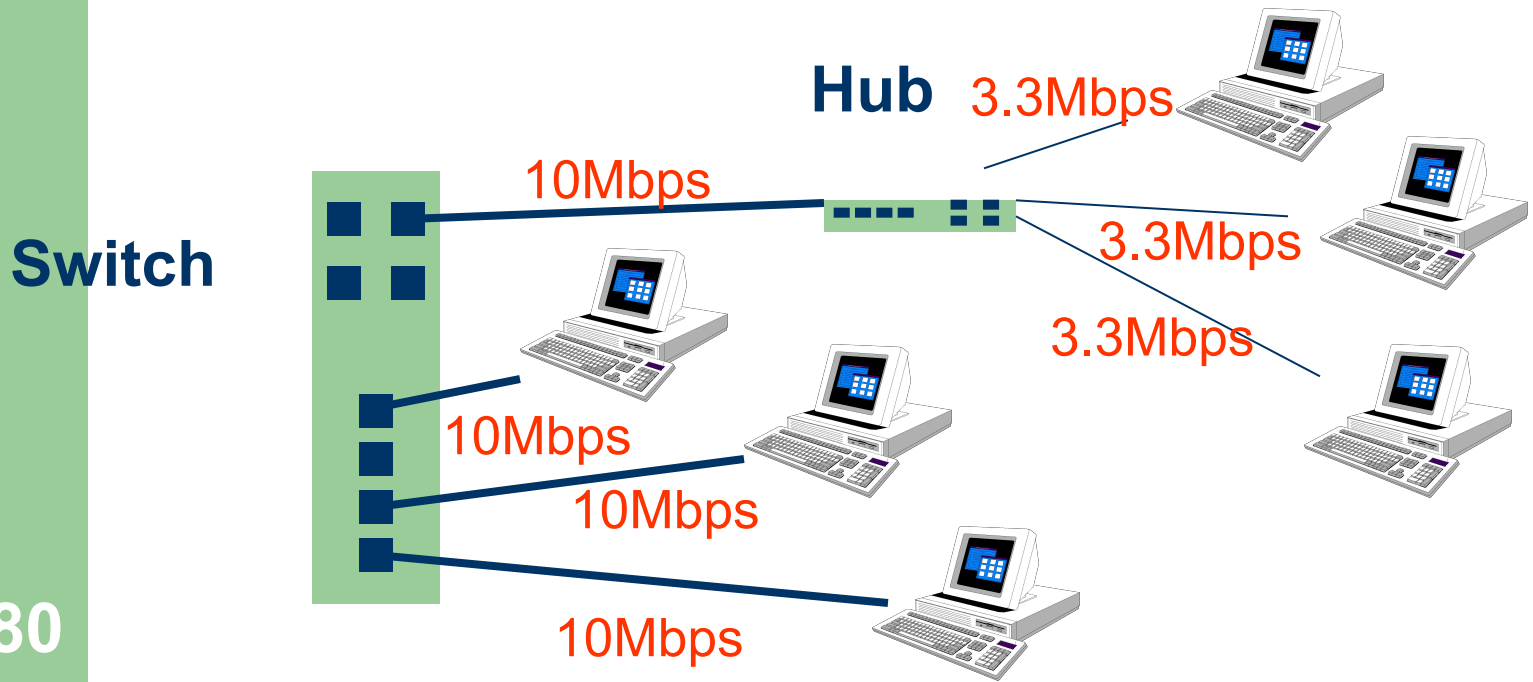
- By having multiports, switches can better use limited bandwidth and prove more cost-effective than bridge



Switches

- Split large networks into small segments, decreasing the number of users sharing the same network resources and bandwidth.
- Understands when two devices want to talk to each other, and gives them a *switched* connection
- Helps prevent data collisions and reduces network congestion, increasing network performance.
- Most home users get very little, if any, advantage from switches, even when sharing a broadband connection.
- Resides on **Layer 2** of the OSI model.

- Switches divide a network into several isolated channels
- Packets sending from 1 channel will not go to another if not specified
- Each channel has its own capacity and need not be shared with other channels



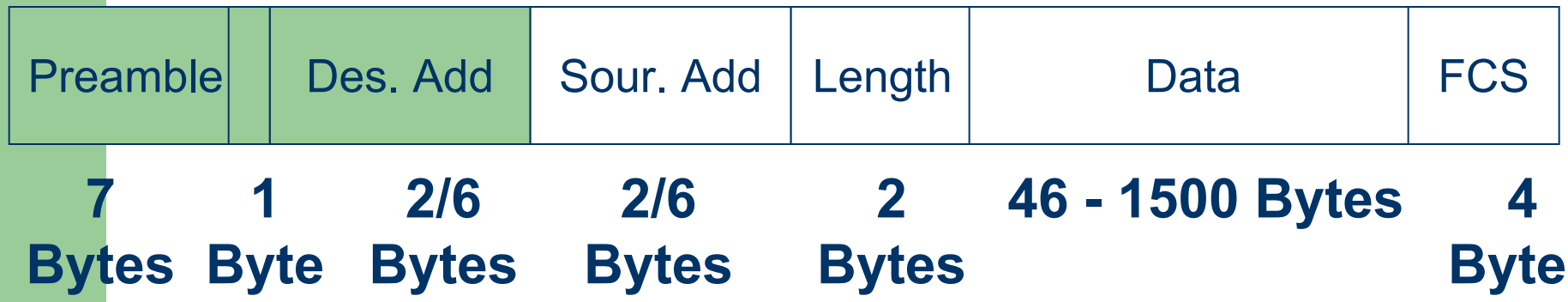
Advantages of Switches

- Switches divide a network into several isolated channels (or collision domains)
  - Reduce the possibility of collision
    - Collision only occurs when two devices try to get access to one channel
    - Can be solved by buffering one of them for later access
  - Each channel has its own network capacity
    - Suitable for real-time applications, e.g. video conferencing
  - Since isolated, hence secure
    - Data will only go to the destination, but not others

Limitations of Switches

- Although contains buffers to accommodate bursts of traffic, can become overwhelmed by heavy traffic
  - Device cannot detect collision when buffer full
    - CSMA/CD scheme will not work since the data channels are isolated, not the case as in Ethernet
  - Some higher level protocols do not detect error
    - E.g. UDP
  - Those data packets are continuously pumped to the switch and introduce more problems

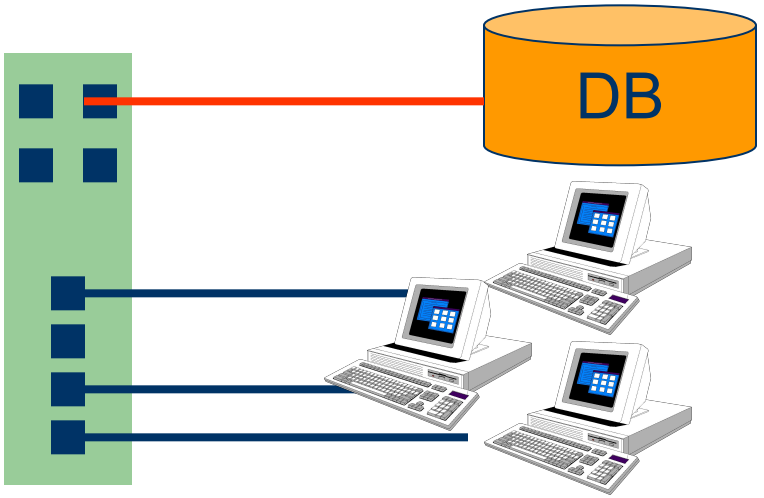
Method of Switching - Cut Through Mode



- Read the first 14 bytes of each packet, then transmit
- Much faster
- Cannot detect corrupt packets
- Can propagate the corrupt packets to the network
- Best suited to small workgroups

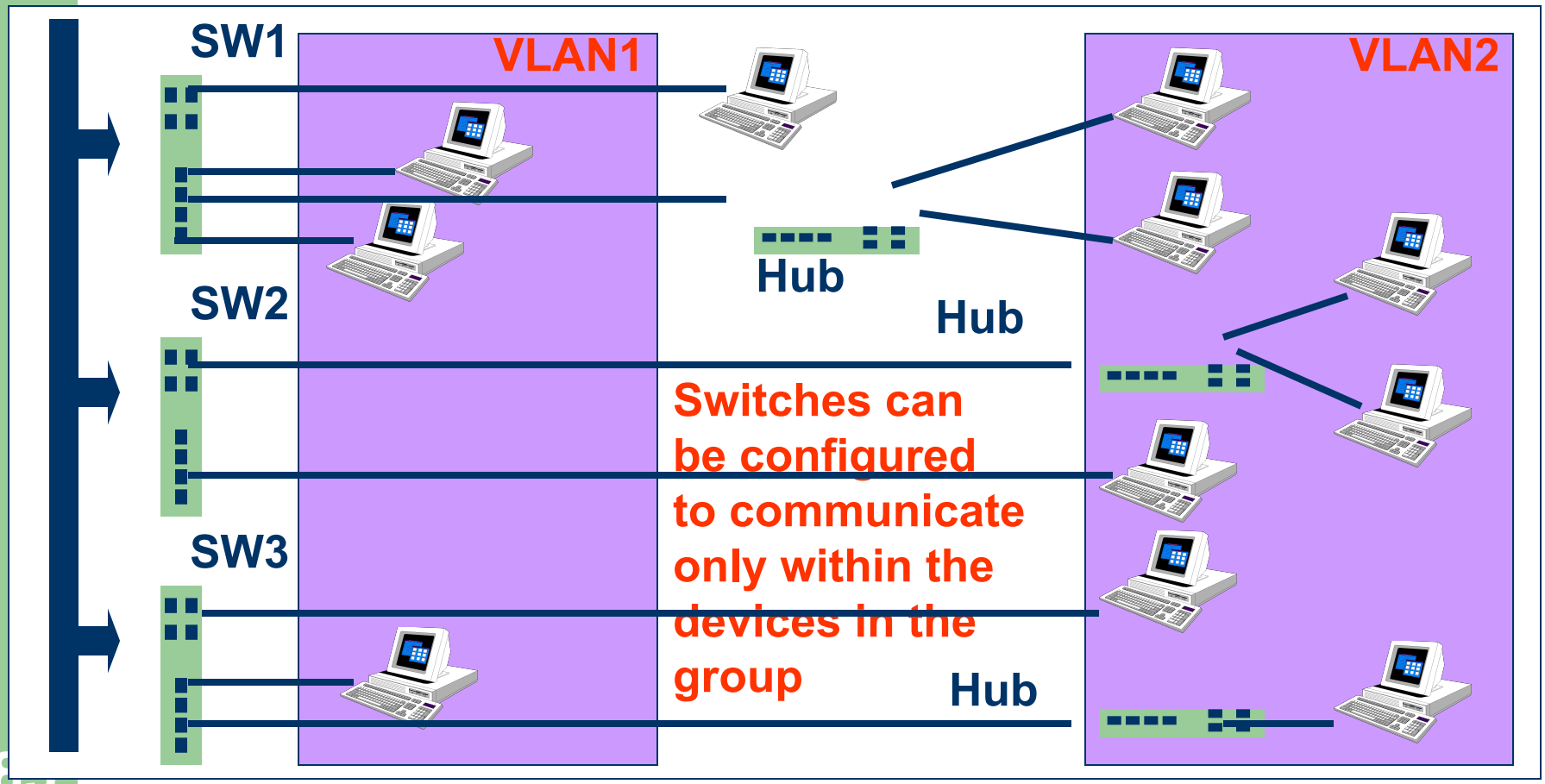
Method of Switching - Store and Forward Mode

- Read the whole packet before transmit
- Slower than the cut-through mode
- More accurate since corrupt packets can be detected using the FCS
- More suit to large LAN since they will not propagate error packets
- Facilitate data transfer between segments of different speed



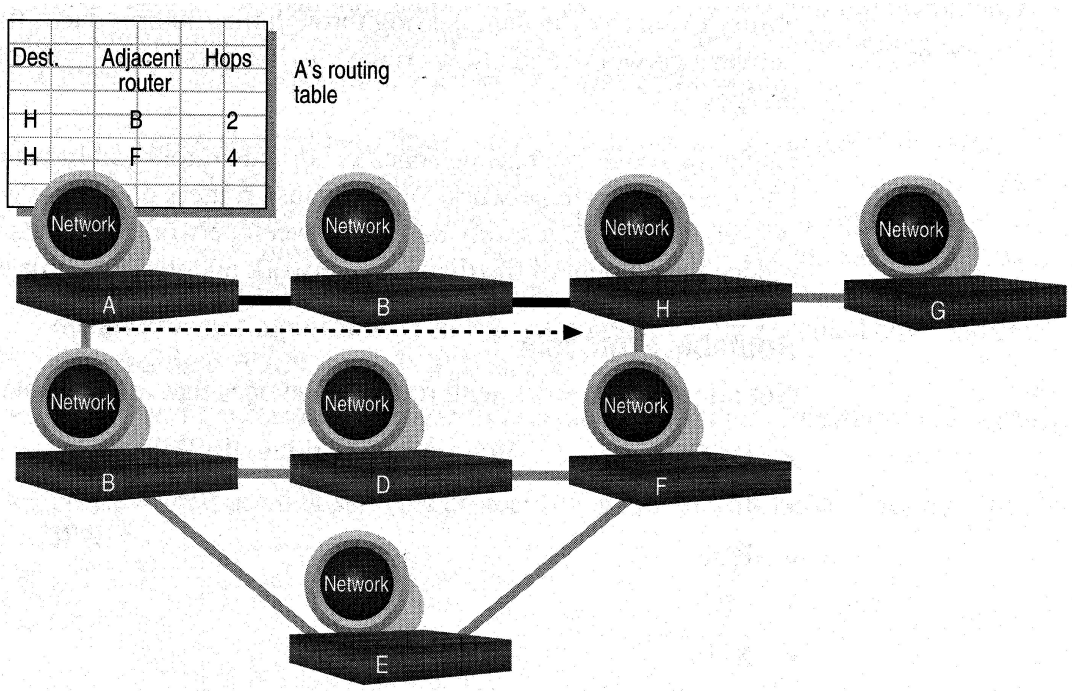
Using Switches to Create VLANs

- Switches can logically group together some ports to form a virtual local area network (VLAN)



d. Routers

- Layer 2 Switches cannot take advantage of multiple paths
- Routers work at the OSI layer 3 (network layer)
- They use the "logical address" of packets and routing tables to determine the best path for data delivery



Router

- A device that connects any number of LANs.
- Uses standardized protocols to move packets efficiently to their destination.
- More sophisticated than bridges, connecting networks of different types (for example, star and token ring)
- Forwards data depending on the Network address (IP), not the Hardware (MAC) address.
- Routers are the only one of these four devices that will allow you to share a single IP address among multiple network clients.
- Resides on Layer 3 of the OSI model.

OSI Model Layers

7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical

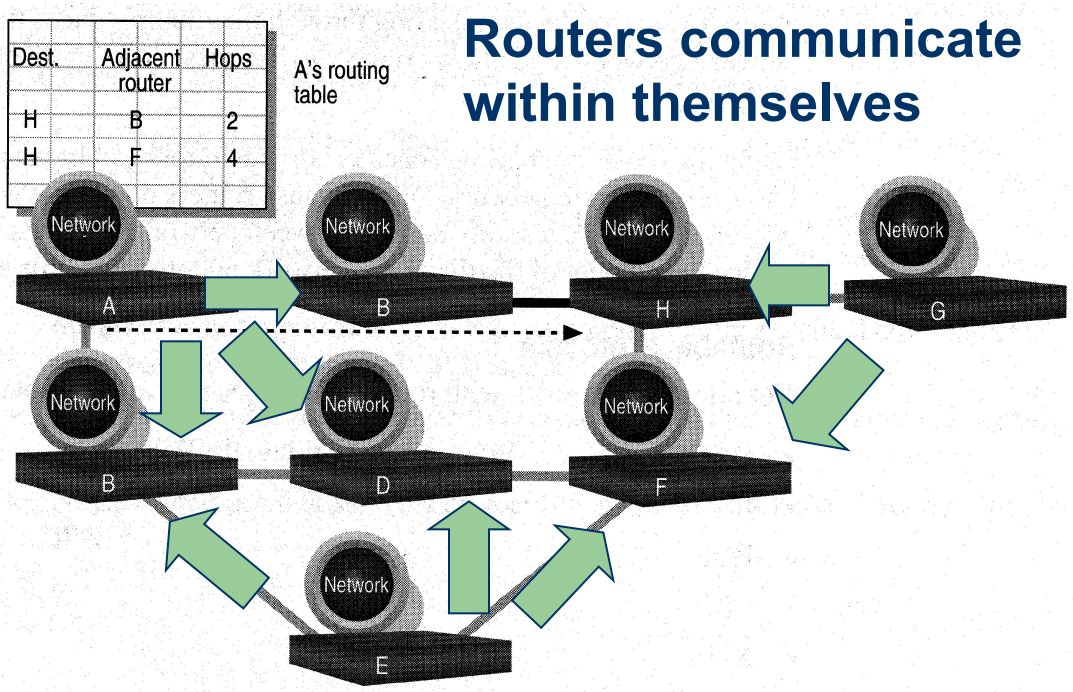


How Routers Work

- As packets are passed from routers to routers, Data Link layer source and destination addresses are stripped off and then recreated
- Enables a router to route a packet from a TCP/IP Ethernet network to a TCP/IP token ring network
- Only packets with known network addresses will be passed - hence reduce traffic
- Routers can listen to a network and identify its busiest part
- Will select the most cost effective path for transmitting packets

How Routing Table is formed

- Routing table is formed based on communications between routers using "Routing Protocols"
  - Routing Protocols ≠ Routable Protocol
- Routing Protocols collect data about current network status and contribute to selection of the best path



Routing Protocol Example - RIP for IP Routing

- RIP (Routing Information Protocol) — the oldest one
- Use no. of hops between nodes to determine best path
- Does not consider the network congestion condition
- Broadcast every 30 sec the routing table to neighbouring routers to convey routing information
- RIP is limited to interpreting a maximum of 16 hops
- Not suitable for large network (e.g. Internet)
- Can create excessive network traffic due to broadcasting
- May take a long time to reach the far reaches

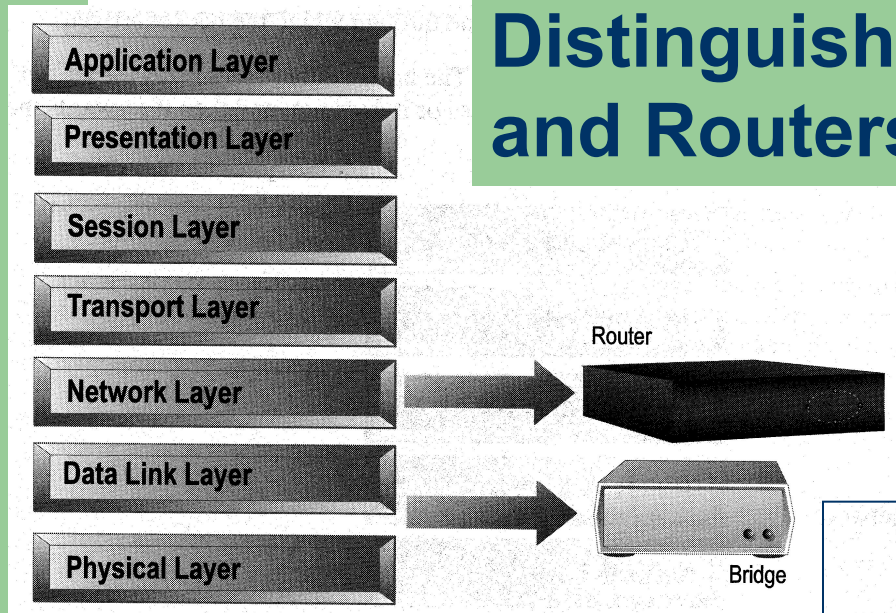
Routing Protocol Example - OSPF for IP

- **OSPF - Open Shortest Path First**
- **Make up the limitations of RIP - can coexist with RIP**
- **In general case, best path refers to the shortest path**
- **In case of traffic congestion, can go a longer path**
- **Each router maintains a database of other router's links**
- **If link failure notice is received, router can rapidly compute an alternate path**
- **Require more memory and CPU power**

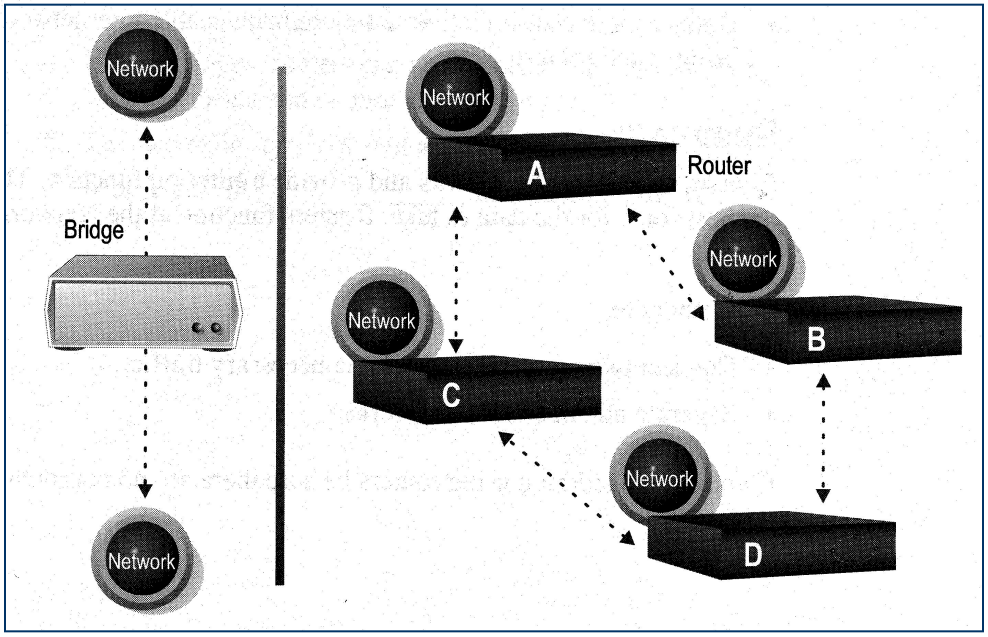
Static and Dynamic Routers

Static Routers	*Dynamic Routers*
Manual configuration of routes	Manual configuration of the first route. Automatic discovery of new routes
Always use the same route	Can select the best route
More secure	Need manual configuration to improve security

Distinguishing Between Bridges and Routers



- **Bridges** forward everything they don't recognize
- **Routers** select the best path



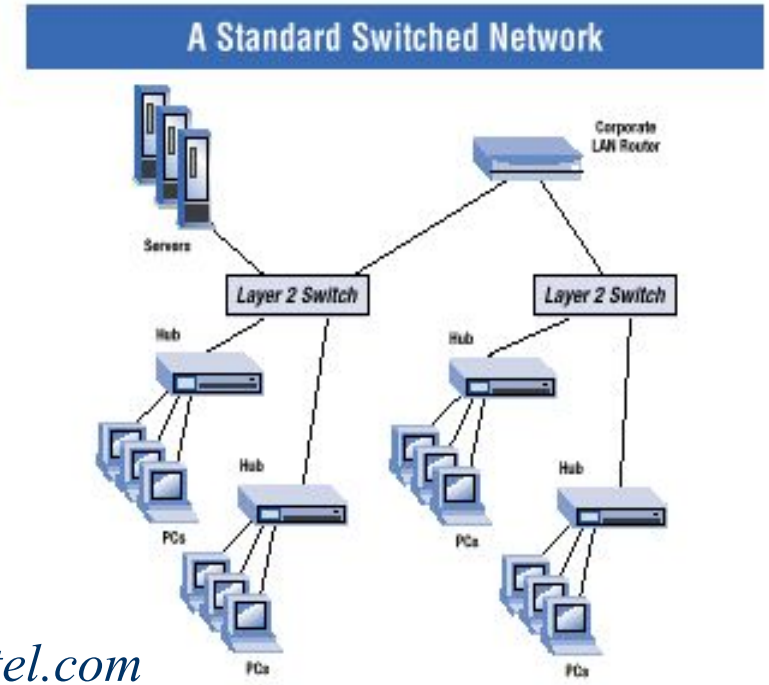
- **Routers** are layer 3 devices which recognize network address
- **Bridges** are layer 2 devices which look at the MAC sublayer node address

Layer-3 Switches

- Layer-3 switches operate in both layer 2 (data link layer) and 3 (network layer)
- Can perform both MAC switching and IP routing
- A combination of switch and router but much faster and easier to configure than router

Why Layer-3 switches?

- Traffic of LAN is no longer local
- Speed of LAN is much faster
- Need a much faster router, however, very expensive



Excerpt from www.intel.com

Summary

- **Repeaters** are the least expensive way to expand a network, but they are limited to connecting two segments
- **Bridges** function similar to repeaters, but can understand the node addresses
- **Switches** can be considered as multiport bridges, can divide a network into some logical channels
- **Routers** interconnect networks and provide filtering functions. They can determine the best route

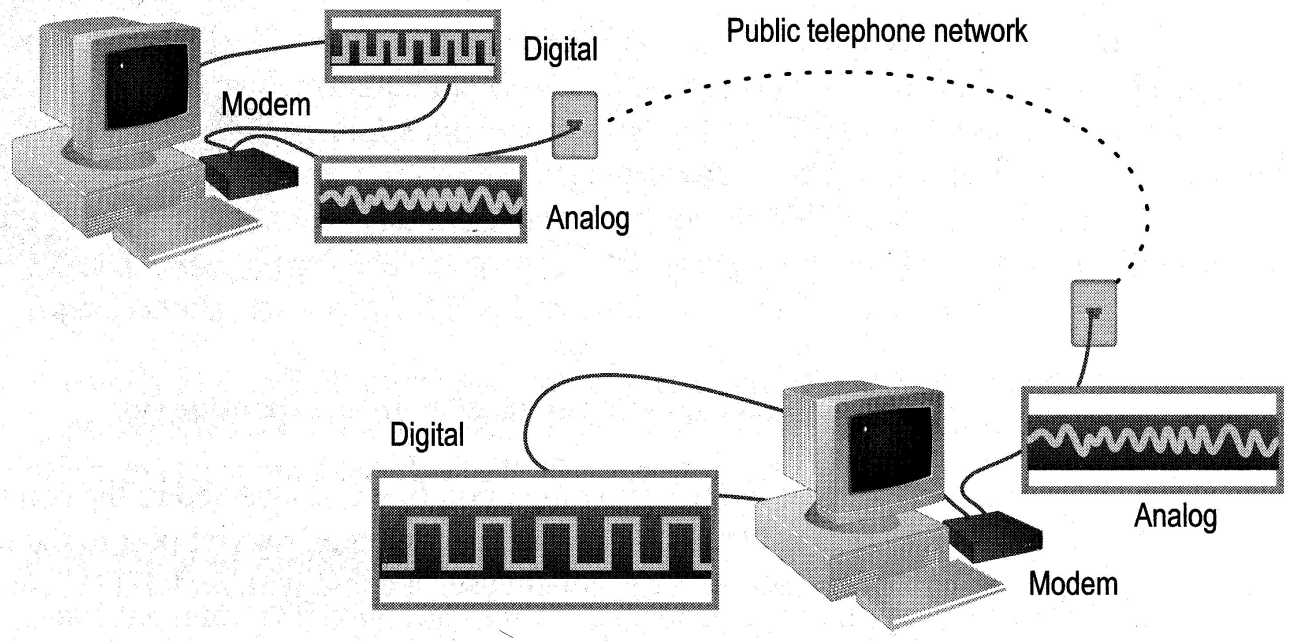
B. Remote Access Devices

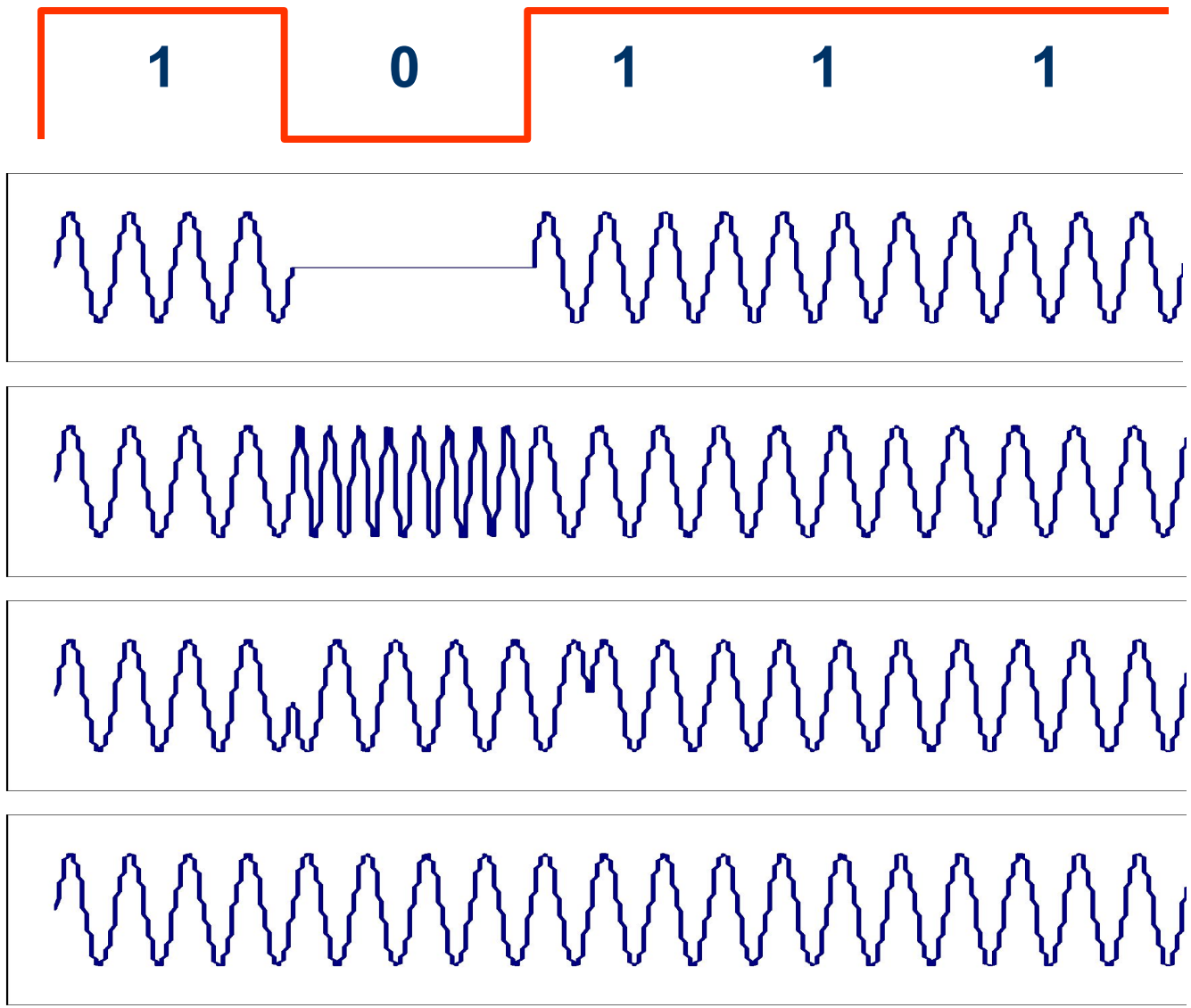
1. Modems



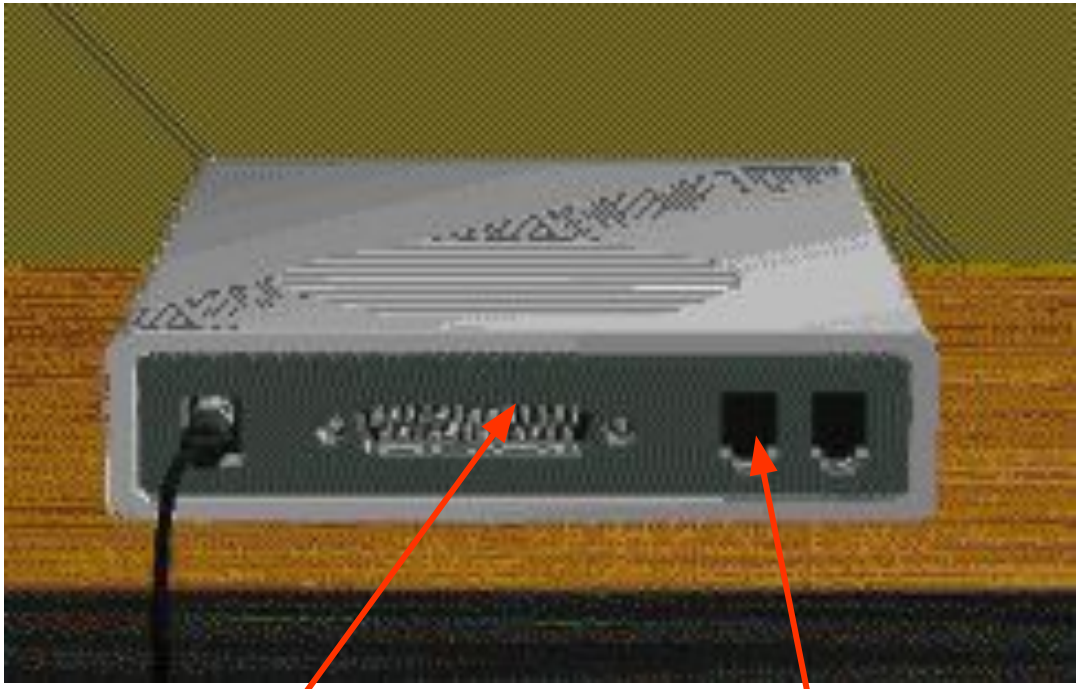
- Allow computers to communicate over a telephone line
- Enable communication between networks or connecting to the world beyond the LAN

- Cannot send digital signal directly to telephone line
- Sending end: MODulate the computer's digital signal into analog signal and transmits
- Receiving end: DEModulate the analog signal back into digital form




1
0
1
1
1
Amplitude Modulation
Frequency Modulation
Phase Modulation
Normal sine wave

- Modems typically have the following I/O interface:
  - A serial RS-232 communication interface
  - An RJ-11 telephone-line interface (a telephone plug)



Modem Standards

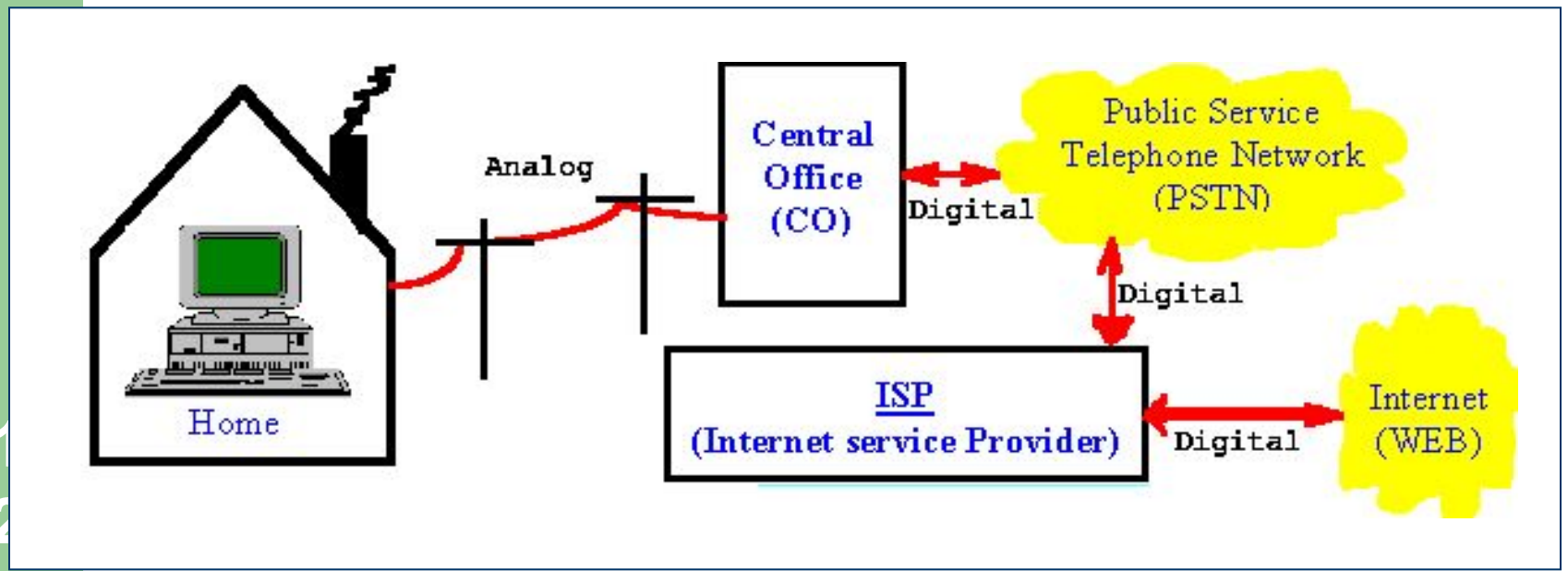
Standard	bps	Introduced	Remarks
V.22bis	2,400	1984	
V.32	9,600	1984	
V.32bis	14,400	1991	
V.32terbo	19,200	1993	Communicate only with another V.32terbo
V.FastClass	28,800	1993	(V.FC)
V.34	28,800	1994	Improved V.FC
V.42bis	115,200	1995	With compression
V.90	56,000	1998	Resolved competition between X2 and Flex56k

Modem Performance Measures

- **Baud rate - the number of symbol change per second on the transmission line**
- **Bit per second (bps) - number of bits transmitted per second**
- **In the past, they are identical**
- **With compression technique, a change of signal can mean more than one bits**
- **28.8kbaud can mean 115.2kbps when using V.42bis**

How V.90 Works

- Modem speed is determined by channel noise level
- The noise level of traditional PSTN (public switch telephone network) limits data rate to ~35kbps
- 56K modem technology assumes only one analog link hence noise level is much lower

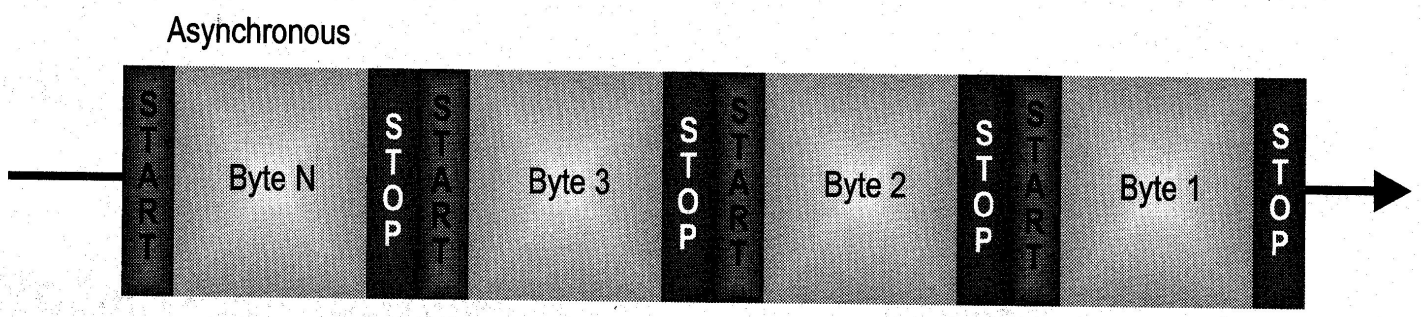


Why V.90 cannot achieve 56kbps in practice?

- The actual data link is 64kbps
- To prevent interference and allow some overhead data in communication, ITU recommends a lower rate to 56 kbps
- However, 56 kbps is a theoretical number
- Depending on the quality and length of the analog link, the actual data rate can range from 30kbps to 53kbps

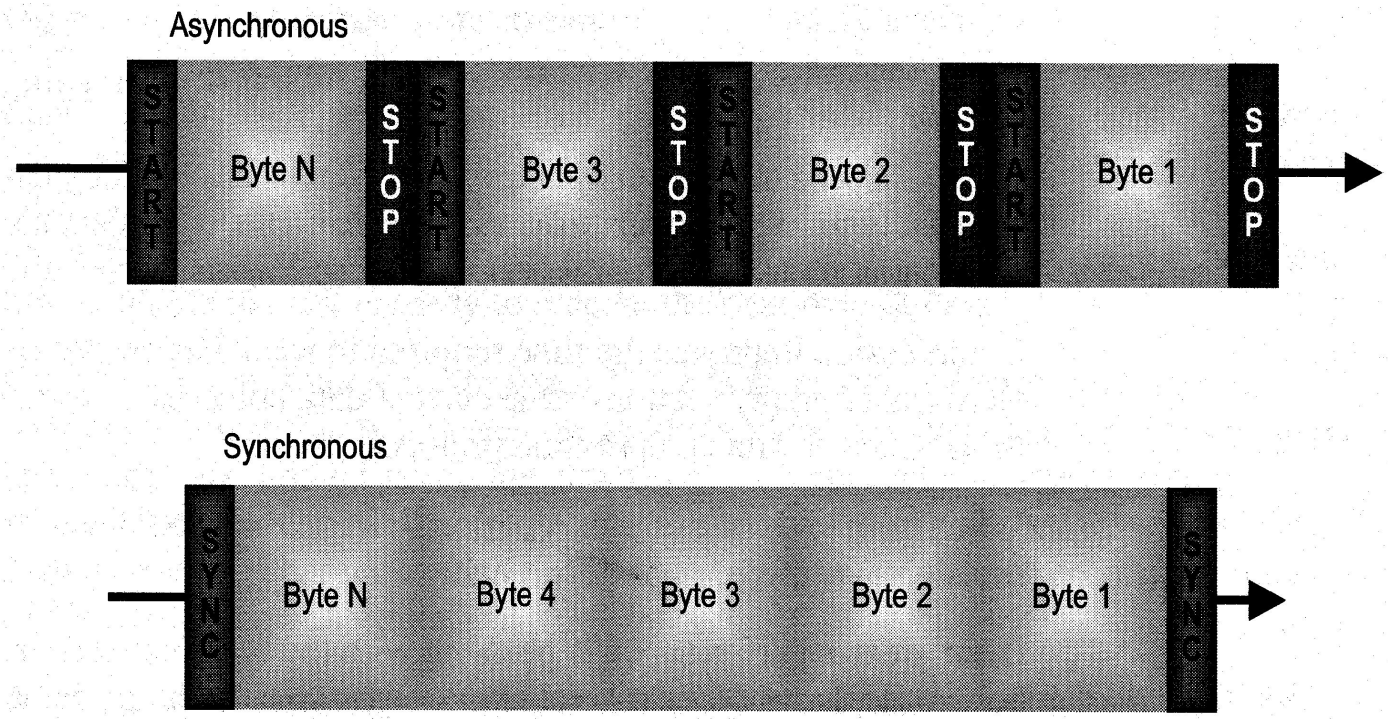
Types of Modem - Asynchronous Modems

- No clocking devices
- Commonly used in telephone networks
- Data is transmitted in a serial stream. Each character is turned into a string of 8 bits
- Each of these characters is separated by one start bit and one or two stop bits



Types of Modem - Synchronous Modems

- Need clocking devices
- Data are transmitted in blocks
- Used in digital networks



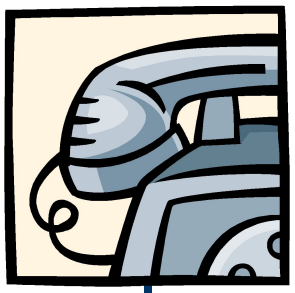
Comparison

- **Asynchronous modems** are relatively simple and economic
  - Large overhead - can be up to 20 to 27% of the data traffic
  - Error control is done by using parity bit or higher layer protocols, e.g. MNP, V.42
- **Synchronous modems** are relatively complicated and expensive
  - Seldom use in home market
  - Less overhead means higher efficiency
  - More sophisticated error control protocol is required

2. ADSL

- ADSL stands for Asymmetric Digital Subscriber Line
- Particularly suitable for high speed multimedia communications, general Internet applications
- Asymmetric - downstream 1.5 to 6.1Mbps
  upstream 16 to 640kbps
- Digital - mainly for transmitting digital data
  still require modulation and demodulation
- Subscriber line - make use of the analog connection between household and CO

ADSL Illustration



normal voice

2 to 3 miles

subscriber line



local loop

Splitter

Telephone Company

low speed



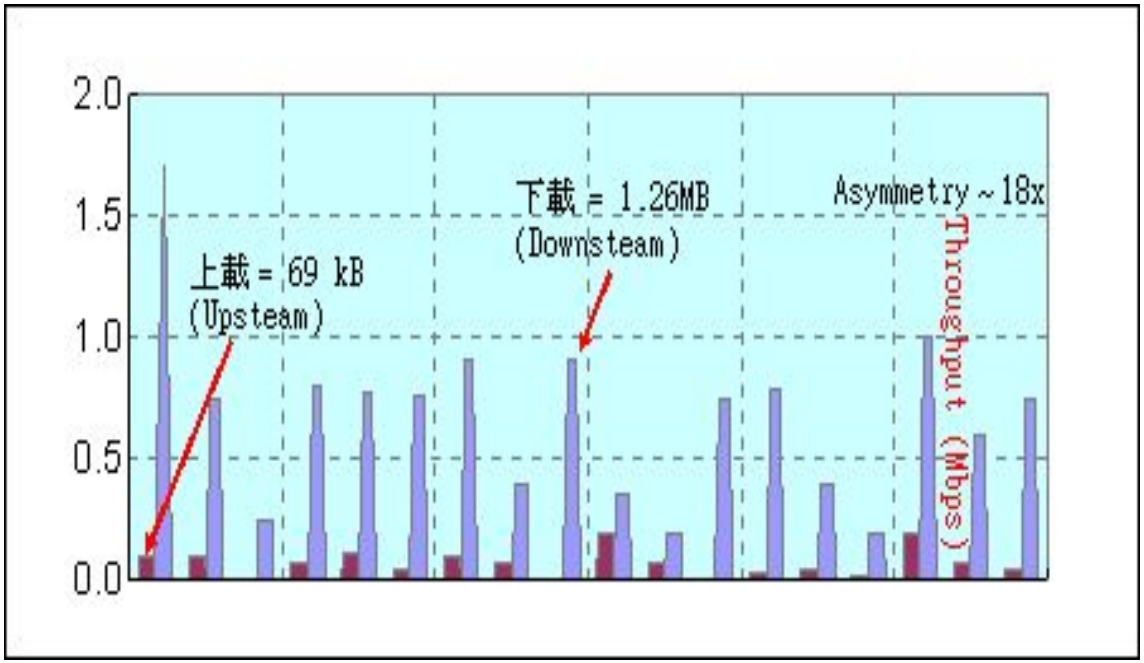
data

high speed

Why Asymmetric?

- In general Internet applications, downstream often requires a higher data rate than upstream
  - Downstream - file download, video playback
  - Upstream - click a link, send a form

- Reducing the resource for upstream can provide more resource for downstream



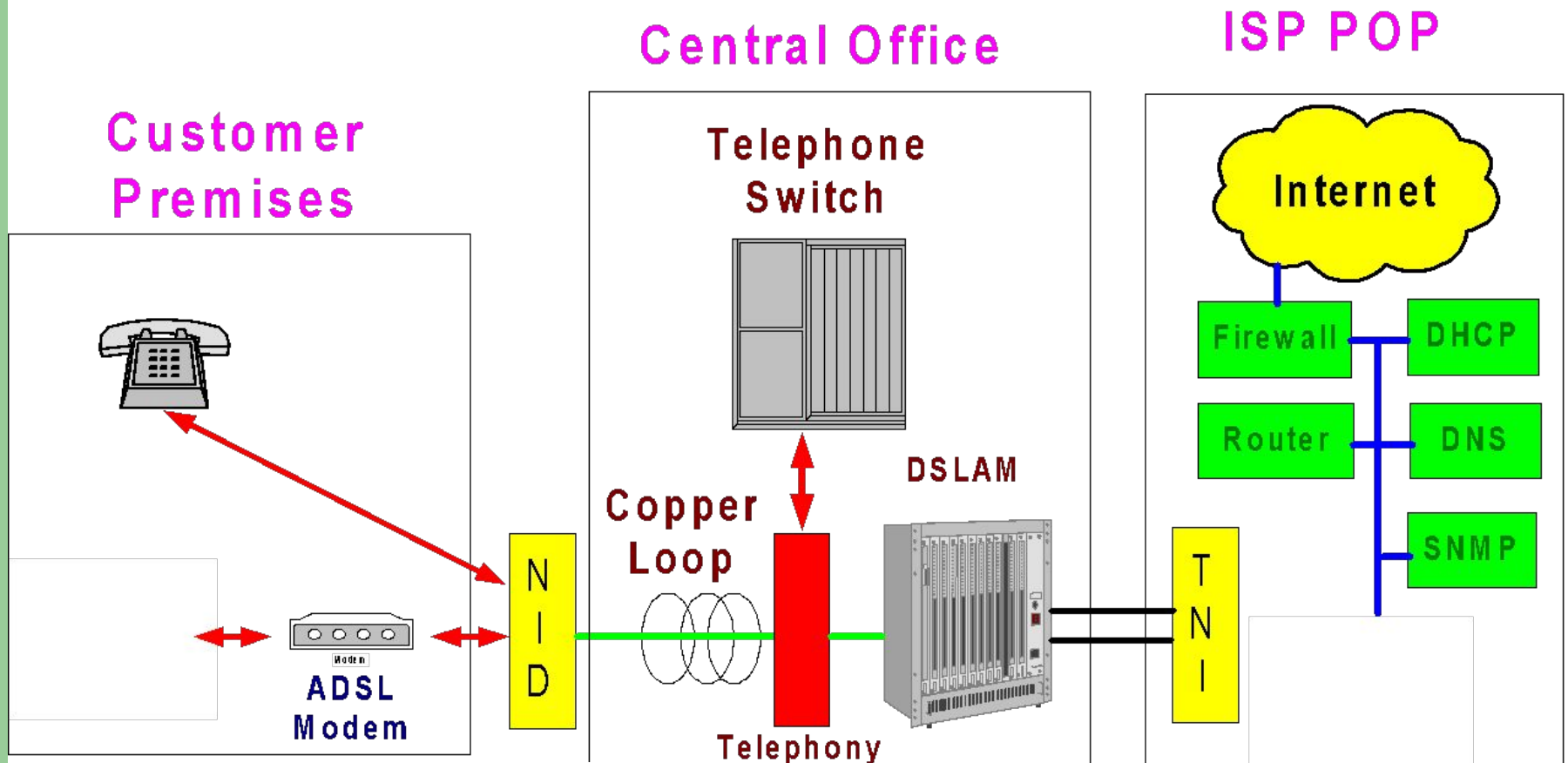
Why Subscriber Line?

- **By better controlling the length and quality of the analog connection between household and CO, a higher data rate can be achieved**

Data Rate	Wire Gauge	Distance	Wire Size	Distance
1.5 or 2 Mbps	24 AWG	18,000 ft	0.5 mm	5.5 km
1.5 or 2 Mbps	26 AWG	15,000 ft	0.4 mm	4.6 km
6.1 Mbps	24 AWG	12,000 ft	0.5 mm	3.7 km
6.1 Mbps	26 AWG	9,000 ft	0.4 mm	2.7 km

- **More than 80% of the current installed subscriber lines can fulfill this requirement**
- **Hence no extra cabling is required**

Architecture of ADSL Services



DSLAM - Digital subscriber line access module (central office ADSL modem pool)

Other DSL Technologies

- **HDSL – High speed DSL**
  - **2 twisted pair, 12,000 feet**
  - **1.5Mbps (DS1) full-duplex**
  - **Symmetric**
- **VDSL – Very high bit rate DSL**
  - **Downstream: 52 Mbps (SONET STS-1) over 1000 feet; or 15 Mbps over 3000 feet**
  - **Upstream: 1.5 to 2.3 Mbps**
- **RDSL – Rate adaptive DSL**
  - **Intelligent DSL to adjust data rate**

FIREWAL
L

Firewalls

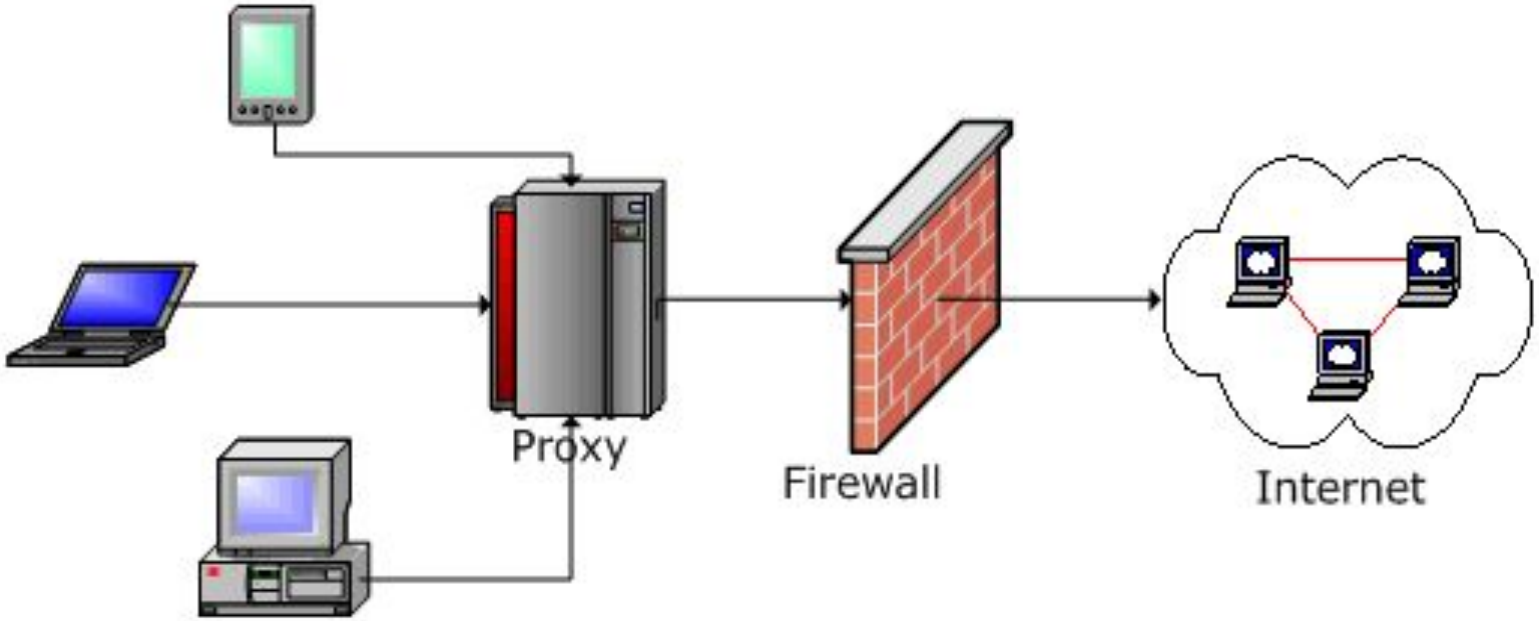
Firewall A Device or Software that separates and protects an Internal Network from a Public Network e.g. the Internet. It prevents and allows some network traffic into and out of the Internal network based on defined rules. In fact it enforces an organization's **Access Security Control Policy.**

- A firewall is a hardware or software

designed to restrict access between or among networks, most often between the Internet and a private Internet.

- The firewall is part of an overall security policy that creates a perimeter defense designed to protect the information resources of the organization.

Firewalls



Firewall filters Network Traffic based on:-

- IP Source Address
- Destination IP Address
- Type of Protocol/Next Header (TCP, UDP, ICMP, etc)
- Source and Destination Ports
- Type of Traffic

FIREWALLS CANNOT

- Protect against attacks that bypass the firewall
  - Dial-out from internal host to an ISP
- Protect against internal threats
  - disgruntled employee
  - Insider cooperates with and external attacker
- Protect against the transfer of virus-infected programs or files
- Protect against Spyware, Key-Loggers, logic Bombs

TYPES OF FIREWALLS

1. PACKET FILTERING FIREWALLS

2. APPLICATION FIREWALLS

3. PROXY SERVERS

PACKET FILTERING FIREWALLS

Data moves inside the Network in form of a PACKET. Packet Filtering Firewalls match packets against a selected criteria to either deny or allow individual data packets. This is done as data packets enter or exit the firewall.

Packet-filtering firewalls validate packets based on protocol, source and/or destination IP addresses, source and/or destination port numbers, time range, type of service (ToS), and various other parameters within the IP Header. Packet filtering is generally accomplished using Access Control Lists (ACL). ACLs are statements configured on the Firewall to allow or deny traffic. Examples of Packet Filtering Firewalls Are **STATELESS** and **STATEFULL** Firewalls.

STATELESS FIREWALLS

Stateless firewalls watch network traffic, and restrict or block packets based on source and destination addresses or other static values like source and destination ports. ***They are not 'aware' of traffic patterns or data flows.*** A stateless firewall uses simple rule-sets that do not account for the possibility that a packet might be fragmented etc.

STATEFUL

Stateful firewalls can watch traffic streams from end to end. They are are aware of communication paths and can implement various IP Security (IPsec) functions such as tunnels and encryption.

In technical terms, this means that stateful firewalls can tell what stage a TCP connection is in (open, open sent, synchronized, synchronization acknowledge or established), it can tell if the MTU has changed, whether packets have fragmented etc.

N.B. Maximum Transmission Unit (**MTU**) is the largest size packet or frame, specified in octets (eight-bit bytes), that can be sent in a packet- or frame-based network such as the Internet.

Application Firewall

An **application firewall** is a form of firewall that controls input, output, and/or access from, to, or by an application or service. It operates by monitoring and potentially blocking the input, output, or system service calls that do not meet the configured policy (criteria) of the firewall. It operates inside the Network and not on the Border. Most common where there is sensitive data.

There are two primary categories of application firewalls, **Network-Based Application Firewalls (Proxy Server)** and **Host-Based Application Firewalls.**

PROXY SERVERS

A **proxy server** is a server that acts as an intermediary for requests from clients seeking resources from other servers.-A client connects to the proxy server, requesting some service, such as a file, connection, web page, or other resource available from a different server and the proxy server evaluates the request as a way to simplify and control its complexity.

Proxy Servers

- Part of an overall Firewall strategy
- Sits between the local network and the external network
  - Originally used primarily as a caching strategy to minimize outgoing URL requests and increase perceived browser performance
  - Primary mission is now to insure anonymity of internal users
    - Still used for caching of frequently requested files
    - Also used for content filtering
- Acts as a go-between, submitting your requests to the external network
  - Requests are translated from your IP address to the Proxy's IP address
  - E-mail addresses of internal users are removed from request headers
  - Cause an actual break in the flow of communications

Security Advantages of Proxy

- Terminates the TCP connection before relaying to target host (in and out)
- Hide internal clients from external network
- Blocking of dangerous URLs
- Filter dangerous content
- Check consistency of retrieved content
- Eliminate need for transport layer routing between networks
- Single point of access, control and logging

Performance Aspects

- Caching
  - By keeping local copies of frequently accessed file the proxy can serve those files back to a requesting browser without going to the external site each time, this dramatically improves the performance seen by the end user
  - Only makes sense to implement this at the ISP rather than the small business level because of the number of pages available
  - Because of dynamic content many pages are invalidated in the cache right away
- Load balancing
  - A proxy can be used in a reverse direction to balance the load amongst a set of identical servers (servers inside the firewall and users outside)
  - Used especially with web dynamic content (.asp, .php,.cfm,.jsp)

Proxy Liabilities

- Single point of failure
  - if the proxy dies , no one can get to the external network
- Client software must usually be designed to use a proxy
- Proxies must exist for each service
- Doesn’t protect the OS
  - proxies run at the application level
- Usually optimized for performance rather than security
  - WINGATE was installed to be easy to configure; opened a winsock proxy to the external interface, which let hackers essentially hijack the machine
- Create a service bottleneck
  - solved via parallelism (more proxies, and load balance)

QUESTION

Compare and Contrast Stateless and Statefull Firewalls. Which one would be suitable in a sensitive environment like a bank. Give reasons for your answer.

Question: Mr Tembo wants to connect two computers which are 10m apart. However, he does not have a switch, bridge, hub or any wireless network device. Advise him on what he needs to do inorder to achieve this goal? Hint: He has an Ethernet Cat6 cable.

PROTOCOLS

A set of guidelines or Language for effecting (implementing networking) communications between **computers**.

EXAMPLES of **protocols** are TCP/IP, UDP, HTTPS, SMTP, DNS etc

A protocol is synonymous with rule. It consists of a set of rules that govern data communications. It determines what is communicated, how it is communicated and when it is communicated. The key elements of a protocol are syntax, semantics and timing

- **Syntax**
- **Semantics**
- **Timing**

Elements of a Protocol

- **Syntax**
  - **Structure or format of the data**
  - **Indicates how to read the bits - field delineation**
- **Semantics**
  - **Interprets the meaning of the bits**
  - **Knows which fields define what action**
- **Timing**
  - **When data should be sent and what**
  - **Speed at which data should be sent or speed at which it is being received.**